

Hardware and Software

Illustrated Slide Deck

A companion visual lesson to HWSW — hardware first (motherboard !' CPU !' cooling !' RAM !' storage !' GPU/AI !' power !' case & ports !' peripherals !' networking), then software (OS !' applications !' cloud, virtualisation, security and AI).

Introduction — what is a computer system?

A computer system is a partnership between two things: the hardware you can touch and the software that tells that hardware what to do. Neither is useful without the other — a laptop with no operating system is a paperweight, and an app with no processor to run on is just a file.

Every machine you will ever support, from a gaming PC to a warehouse scanner to a phone, follows the same four-part pattern: input, processing, storage, and output. Once you can spot those four parts, you can reason about any device.

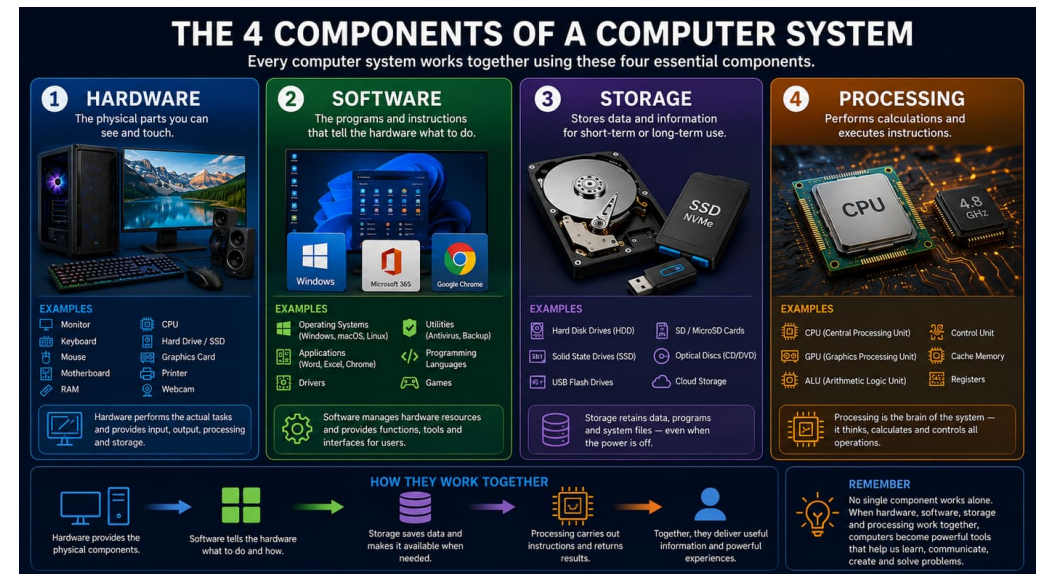
In Part 1 we open the case and work through the physical components — motherboard, CPU, cooling, RAM, storage, power supply, ports, peripherals, and the networking hardware that puts a device online.

In Part 2 we move up the stack into software — firmware, the operating system, the applications people use every day, and the modern cloud, virtualisation, security and AI layers that sit on top.

By the end of the deck you should be able to point at any part of a computer, name it, explain what it does, and know whether it lives on the hardware side or the software side.

The four components of a computer system — the map for the whole lesson

- Every computer = input + processing + storage + output.
- Hardware is what you touch; software is what runs on it.
- Use this map to place every other slide in the deck.

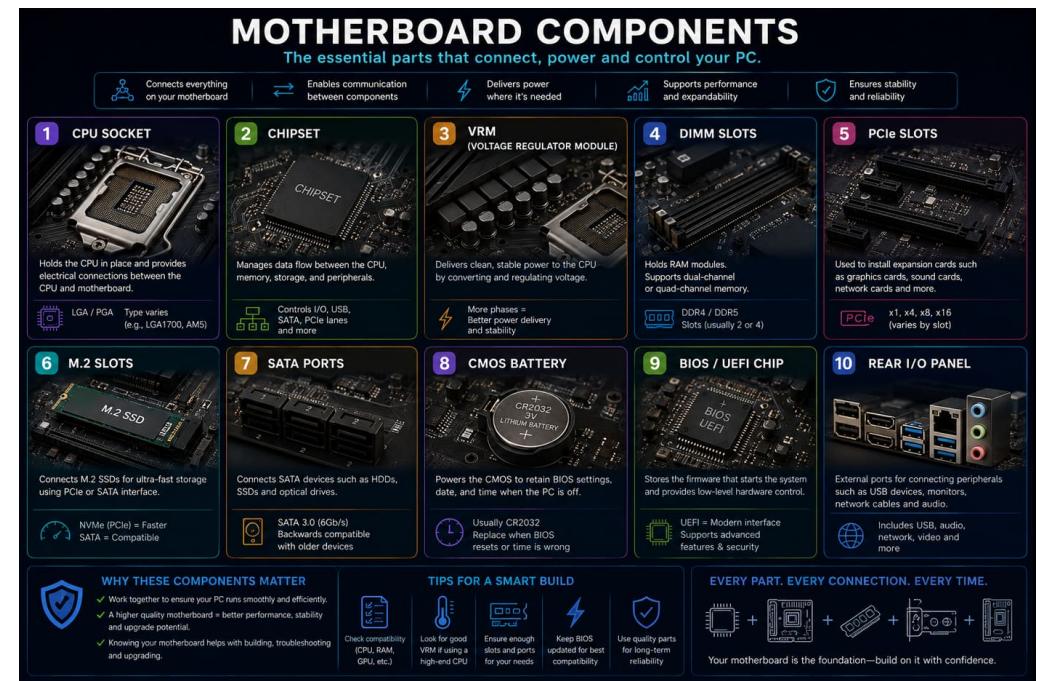


PART 1 — HARDWARE: the physical machine

Every slide in Part 1 shows a physical component you can point at inside a PC case. Work outwards from the motherboard: processors, cooling, memory, storage, power, case & ports, input and output devices, and networking hardware.

Motherboard components — the labelled overview

- Central hub that connects CPU, RAM, storage, GPU and PSU.
- Learn the slot names: CPU socket, DIMM, PCIe, M.2, SATA.
- Chipset routes traffic between all components.



The CPU socket — where the processor lives

- Physical mount that connects the CPU to the board.
- Socket type (LGA, PGA, BGA) must match the CPU.
- Damaged pins here = dead motherboard.

MOTHERBOARD COMPONENTS

CPU SOCKET

The CPU socket is the interface on the motherboard that holds the processor in place and provides electrical connections between the CPU and the motherboard.

WHAT IS A CPU SOCKET?

It is a physical and electrical connection point that allows the CPU to communicate with the motherboard. Different CPU generations use different socket types.

KEY FUNCTIONS

- Holds the CPU securely in place
- Provides electrical contact for data, power, and control signals
- Ensures proper alignment and installation
- Determines CPU compatibility with the motherboard

CPU SOCKET ANATOMY (INTEL LGA EXAMPLE)

- Load Plate:** Presses down to secure the CPU in place
- Retention Arm:** Locks the CPU in place when closed
- Alignment Notches:** Ensure the CPU is installed in the correct orientation
- Contact Pads / Pins (land grid array):** Provide electrical connections between the CPU and motherboard
- Socket Frame:** Holds all components together and aligns the CPU

COMMON CPU SOCKET TYPES

INTEL SOCKETS (LGA – Land Grid Array)

LGA 1700 12th, 13th, 14th Gen Core Processors	LGA 1200 10th, 11th Gen Core Processors	LGA 1151 6th, 7th, 8th, 9th Gen Core Processors	LGA 2066 Intel Core X-Series (High End)
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AMD SOCKETS (PGA – Pin Grid Array / LGA)

AM5 AMD Ryzen 7000 Series	AM4 AMD Ryzen 1000 to 5000 Series	AM3+ AMD FX / Ryzen (Older)	TR4 / sTRX4 AMD Threadripper Series
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CPU SOCKET COMPATIBILITY

The CPU socket determines which processors are compatible with your motherboard. Always check the motherboard manual or vendor website for supported CPUs.

SOCKET TYPE	CHIPSET	BIOS VERSION
Must match the CPU	Determines features and compatibility	May need update to support new CPUs

CPU SOCKET INSTALLATION (LGA EXAMPLE)

1. Open the retention arm and lift the load plate.
2. Align the CPU notches with the socket.
3. Place the CPU gently into the socket.
4. Close the load plate and lock the retention arm.

LGA vs PGA SOCKETS

LGA (Land Grid Array)	PGA (Pin Grid Array)
✓ Pins are on the motherboard ✓ More durable (CPU has flat pads) ✓ Better for high pin counts ✓ Typical for Intel (modern)	✗ Pins are on the CPU ✗ More fragile (pins can bend) ✗ Older design ✗ Typical for AMD (older)

TIPS & BEST PRACTICES

- Always power off and unplug before installing a CPU.
- Handle the CPU by the edges. Avoid touching the contacts.
- Ensure proper alignment using the notches or keys.
- Do not force the CPU into the socket.
- Apply thermal paste and install a cooler after installation.

DID YOU KNOW?

The "LGA" in Intel sockets stands for Land Grid Array, whereas "PGA" in older AMD sockets stands for Pin Grid Array.

GLOSSARY

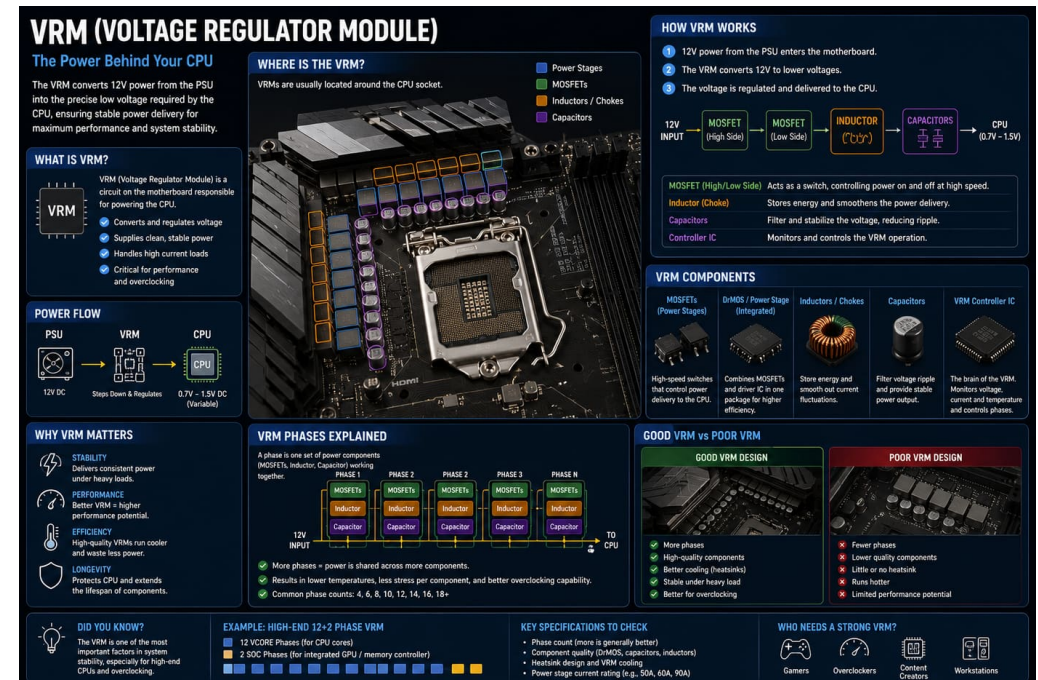
LGA: Land Grid Array – Flat contacts on the CPU that land on pads in the socket.
 PGA: Pin Grid Array – Pins on the CPU that go into holes in the socket.
 VRM: Voltage Regulator Module – Delivers power to the CPU.
 Chipset: Manages communication between CPU and other components.

COMPATIBILITY CHECK

Use tools like your motherboard manufacturer's website or PCPartPicker to verify CPU compatibility.

VRM (Voltage Regulator Module) — the power behind your CPU

- Steps 12 V from the PSU down to ~1 V for the CPU.
- Weak VRM = instability under heavy load.
- Runs hot — needs its own heatsink on gaming boards.



BIOS / UEFI chip — the firmware that starts your PC

- First code to run when you press power.
- Initialises hardware, then hands over to the OS.
- UEFI replaces the older BIOS with a modern interface.

BIOS/UEFI CHIP

THE FIRMWARE THAT STARTS YOUR PC

The BIOS/UEFI chip is a small memory chip on the motherboard that stores the firmware (software) used to start and initialize your computer before the operating system loads.

KEY FUNCTIONS

- Starts the System**
Executes Power-On Self Test (POST) and initializes hardware.
- Configures Hardware**
Sets up and configures all connected hardware components.
- Boots the OS**
Loads the bootloader from the selected device to start the operating system.
- Provides Setup Interface**
Stores settings like boot order, date/time, fan control, and security options.
- Firmware Updates**
Can be updated to add features, improve compatibility, and fix issues.

BIOS is traditional firmware using a legacy interface. UEFI (Unified Extensible Firmware Interface) is the modern replacement with more features, faster boot, and support for large drives and secure boot.

HOW IT WORKS (BOOT PROCESS)

- Power On**
You press the power button.
- BIOS/UEFI Executes**
The firmware in the chip starts and runs POST.
- Hardware Initialization**
The firmware detects and initializes all hardware.
- Boot Device Selected**
It looks for the boot device based on the boot order.
- OS Loads**
The bootloader is loaded and the operating system starts.

HOW TO UPDATE BIOS/UEFI

- Prepare OS**
Connect BIOS file from the motherboard manufacturer.
- Copy the file to a USB drive (FAT32 formatted).**
- Enter BIOS/UEFI**
Boot into BIOS and go to the update utility.
- Select the file and start the update.**
- Wait for completion and reset the system.**

Do not turn off the power during a BIOS/UEFI update! It can damage the chip.

DID YOU KNOW?

The BIOS/UEFI chip can be reprogrammed or replaced if the firmware becomes corrupted.

CHIP REMOVAL IF NEEDED

You can remove the chip with a chip puller or hot air gun and program it using an external SPI programmer.

BIOS RECOVERY OPTIONS

- Dual BIOS / Backup BIOS
- BIOS Flashback Button
- External BIOS Programmer

IMPORTANT!

Flashing the wrong BIOS or interrupting the update can make your motherboard unusable (bricked). Always follow the manufacturer's instructions carefully.

RELATED COMPONENTS

- CMOS Battery
- Chipset
- SPI Flash Chip
- Clear CMOS Jumper

COMMON BIOS/UEFI CHIP TYPES

Manufacturer	Model	Capacity
Winbond	W25Q128LV	128 Mbit
Macronix	MX25U1280	128 Mbit
GigaDevice	GD25Q128	128 Mbit
ISSI	IS25SL128	128 Mbit

BIOS vs UEFI

Feature	BIOS (Legacy)	UEFI (Modern)
Interface	16-bit	32/64-bit
Boot Speed	Slower	Faster
Drive Support	Up to 2.2 TB (MBR)	Over 2.2 TB (GPT)
Security	Basic	Secure Boot, TPM
User Interface	Text-based	Graphical (GUI)
Mouse Support	No	Yes
Network Boot	Limited	Yes

WHERE IS IT LOCATED?

The BIOS/UEFI chip is typically located:

- Near the chipset
- Near the CMOS battery
- Near PCIe slots

It is usually a small 8-pin or 16-pin chip labeled with the manufacturer name.

COMMON BIOS/UEFI CHIP SOCKETS

- SOIC-8 (8-Pin)**
Common for smaller motherboards.
- SOIC-16 (16-Pin)**
Most common for desktop motherboards.
- WSON-8 / DFN**
Used on some modern motherboards.

SOME POPULAR MANUFACTURERS

- AMI (American Megatrends)
- Insyde
- Phoenix
- Coreboot
- Tianocore

CMOS battery — keeps BIOS settings and the clock alive

- Small coin cell (CR2032) on the motherboard.
- Keeps date, time and BIOS settings when unplugged.
- Dead battery = clock resets, boot errors.

CMOS BATTERY

THE POWER SOURCE FOR BIOS SETTINGS

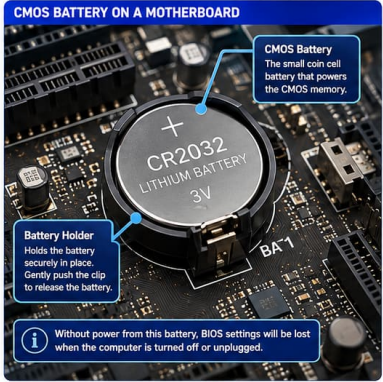
The CMOS battery is a small coin cell battery on the motherboard that powers the CMOS memory. It keeps your BIOS settings, date, and time saved when the computer is turned off or unplugged.

CMOS stands for Complementary Metal-Oxide-Semiconductor

KEY FUNCTIONS

- Maintains BIOS Settings**
Keeps BIOS configuration, boot order, and preferences saved.
- Keeps Date & Time**
Powers the real-time clock (RTC) when the system is off.
- Stores Hardware Information**
Preserves system hardware settings and configurations.
- Enables System Startup**
Ensures the system can start correctly with saved settings.

CMOS BATTERY ON A MOTHERBOARD

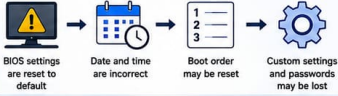


COMMON CMOS BATTERY TYPE



- ✓ Type: CR2032 (most common)
- ✓ Voltage: 3 Volts
- ✓ Chemistry: Lithium
- ✓ Diameter: 20 mm
- ✓ Thickness: 3.2 mm

WHAT HAPPENS WHEN THE BATTERY IS DEAD?



HOW TO IDENTIFY A LOW OR DEAD CMOS BATTERY

- BIOS settings keep resetting after saving
- Incorrect date and time on every boot
- "CMOS Checksum Error" or "CMOS Battery Low" message
- System asks to press F1/F2 to continue

CMOS Checksum Error Defaults Loaded Press F1 to Continue

HOW TO REMOVE & REPLACE

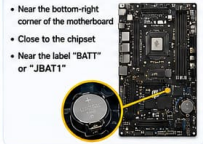
1. Power off and unplug the computer.
2. Open the case and locate the CMOS battery.
3. Gently push the retention clip outward.
4. Remove the old battery from the holder.
5. Insert the new battery (positive side "+" facing up).
6. Press it down until it clicks into place.

Always handle the battery by the edges. Avoid touching the metal surfaces.

WHERE IS IT LOCATED?

The CMOS battery is typically found:

- Near the bottom-right corner of the motherboard
- Close to the chipset
- Near the label "BATT" or "JBAT1"



TIPS & BEST PRACTICES

- ✓ Replace the battery every 2-5 years.
- ✓ Use a high-quality CR2032 battery from a trusted brand.
- ✓ Ensure the "+" side is facing up when installing.
- ✓ After replacing, you may need to reconfigure BIOS settings.
- ✓ Dispose of old batteries properly. Do not recharge or burn.

DID YOU KNOW?

A CMOS battery can last 2 to 5 years depending on usage and environment.

LITHIUM BATTERY SAFETY

- Swallowing lithium batteries can be extremely dangerous.
- Keep out of reach of children.
- Seek medical attention immediately if swallowed.

SYMPTOMS vs SOLUTION

Symptoms	Solution
• BIOS keeps resetting	• Replace the CMOS battery
• Wrong date and time	• Re-enter BIOS settings
• CMOS error message	• Save and exit

COMPATIBILITY NOTE

Most motherboards use CR2032, but some may use CR2025 or other types. Always check your motherboard manual.

PRO TIP

If you remove the CMOS battery, the system may lose power to CMOS memory. If replacing the motherboard, reinstall the battery before powering on if preconfigured.

CPU (Central Processing Unit) — anatomy of the chip

- Executes the instructions of every program.
- Billions of transistors; multiple cores and threads.
- Clock speed × cores × cache = real-world performance.

02
CPU (Central Processing Unit)
The Brain of the Computer

The CPU is the most important component in your computer. It thinks, calculates, and gives instructions to make everything work.

What is it?

The CPU (Central Processing Unit) is a tiny chip that performs all the calculations and runs the instructions (programs) on your computer.

- Executes instructions
- Performs calculations
- Controls all operations
- The "brain" of the system

How it works

The CPU fetches instructions, decodes them, processes the data, and returns the results.

- Fetch**: Gets the next instruction from memory.
- Decode**: Understands what the instruction means.
- Execute**: Performs the calculation or task.
- Write Back**: Saves the result back to memory or sends it to another part.

Where it is located

The CPU sits in the CPU socket on the motherboard under a cooler (fan or liquid cooler).

Inside the CPU

Control Unit: Tells the CPU what to do.

ALU (Arithmetic Logic Unit): Performs math and logic operations.

Registers: Tiny ultra-fast storage for data the CPU is using right now.

Cache (L1, L2, L3): Very fast memory that stores data the CPU uses often.

Cores: Each core can work on tasks. More cores = better multitasking.

Key Specifications

- Cores**: Number of independent processing units. More cores = better multitasking.
- Threads**: Each core can handle multiple threads. More threads = smoother performance.
- Clock Speed (GHz)**: How fast a core can work. Higher GHz = faster (but not everything!). More cache = less waiting.
- TDP (Thermal Design Power)**: Amount of heat (in watts) the CPU produces. Lower TDP = cooler and more efficient.

Popular Modern Examples

- Intel Core i9**: Intel Core i9-14900K. High performance for gaming, creation & work.
- AMD Ryzen 9**: AMD Ryzen 9 7950X. Excellent for multitasking & productivity.
- Apple M3 Pro**: Apple M3 Pro. Powerful & efficient for Mac computers.

Real-World Uses

- Running apps (Word, Chrome, Photoshop)
- Playing games
- Editing videos and photos
- Programming and compiling code
- 3D modeling and animation
- Artificial Intelligence & Machine Learning

Did You Know?

Modern CPUs have billions of transistors!

A tiny CPU can do billions of calculations every second.

Knowledge Check

What is the main job of the CPU?

- Store data
- Display images
- Process data and run instructions
- Provide power to the computer

Analogy

CPU = Brain
Motherboard = Body
RAM = Desk (workspace)
Storage = Bookshelf
GPU = Artist
PSU = Power from food

KEY TAKEAWAY:

The CPU is the brain of your computer. It processes instructions, performs calculations, and controls everything. A better CPU means a faster and smarter computer!

CPU cache — L1, L2 and L3

- Tiny, extremely fast memory next to the CPU cores.
- L1 fastest/smallest, L3 largest/shared.
- Big cache helps games, databases and AI a lot.

CPU CACHE (L1, L2, L3)

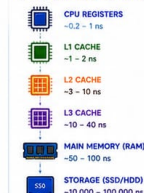
The Memory Hierarchy Inside the CPU

CPU cache is a small, ultra-fast memory located on or very close to the CPU cores. It stores copies of data and instructions that the CPU uses most often, so they can be accessed much faster than from RAM.

WHY DO WE NEED CACHE?

The CPU is extremely fast, but main memory (RAM) is much slower. Cache acts as a high-speed buffer between the CPU and RAM, reducing the average time to access data.

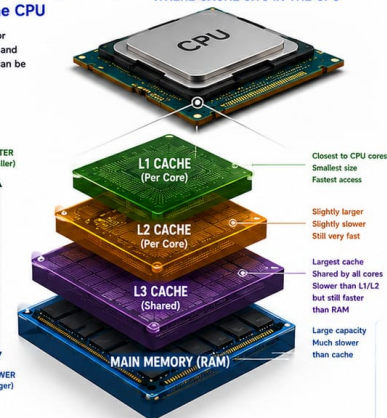
MEMORY HIERARCHY (FROM FASTEST TO SLOWEST)



FASTER
(Smaller)

SLOWER
(Larger)

WHERE CACHE SITS IN THE CPU



CACHE LEVELS EXPLAINED

L1 CACHE (LEVEL 1)

- Smallest and fastest cache
- Usually split into:
 - L1 Instruction Cache (I-Cache)
 - L1 Data Cache (D-Cache)
- Private to each CPU core
- Size: 16 KB - 128 KB per core (typical)
- Latency: ~1 - 2 nanoseconds

Holds:

- Frequently used instructions
- Frequently used data

L2 CACHE (LEVEL 2)

- Larger than L1, slightly slower
- Private to each CPU core
- Size: 256 KB - 2 MB per core (typical)
- Latency: ~3 - 10 nanoseconds

Holds:

- Less frequently used instructions and data from L1 and RAM

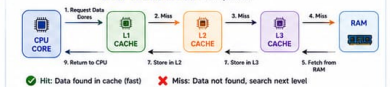
L3 CACHE (LEVEL 3)

- Largest cache, shared by all cores
- Acts as a last-level cache before going to RAM
- Size: 4 MB - 64 MB+ (shared)
- Latency: ~10 - 40 nanoseconds

Holds:

- Data and instructions shared across all cores

HOW CACHE WORKS (Simplified)



KEY BENEFITS OF CACHE

- SPEED:** Cache is much faster than RAM, reducing access time significantly.
- PERFORMANCE:** More data supplied to the CPU per second = better overall performance.
- REDUCED LATENCY:** Lower average time to access data and instructions.
- EFFICIENT MEMORY USE:** Stores only what's needed most often, saving bandwidth.



NOTE: Cache sizes and latencies vary between CPU architectures and generations. Larger cache generally improves performance, but is more expensive and uses more chip area and power.

TYPICAL CACHE SIZES (EXAMPLES)

CPU EXAMPLE	L1 CACHE (Per Core) I-Cache	L1 CACHE (Per Core) D-Cache	L2 CACHE (Per Core)	L3 CACHE (Shared)	CORES
Intel Core i5 (12 th Gen)	32 KB	48 KB	1.25 MB	18 MB	6
AMD Ryzen 5 5600X	32 KB	32 KB	512 KB	32 MB	6
Apple M2 (Performance Core)	128 KB	128 KB	4 MB	16 MB	8
Intel Core i9 (14 th Gen)	32 KB	48 KB	2 MB	36 MB	24

Different CPUs and GPUs — how modern processors compare

- Intel vs AMD vs Apple: different sockets, same job.
- NVIDIA vs AMD vs Intel Arc for GPUs.
- Pick the chip that matches the user's workload.

DIFFERENT CPUs AND GPUs
THE TWO BRAINS OF MODERN COMPUTERS

CPUs handle general tasks and run your operating system and applications.
GPUs handle graphics, gaming, AI, and massively parallel workloads.

SIMPLE IDEA
The CPU is like the brain of your computer. It processes information, makes decisions, and controls all the other parts.

CENTRAL PROCESSING UNITS (CPUs)

DESKTOP CPUs For desktop computers and workstations

Intel Core i9-14900K	Intel Core i5-14400	AMD Ryzen 9 7950X	AMD Ryzen 5 7600	Intel Core Ultra 7 155H
High performance for enthusiasts and gamers	Great balance of performance and price	Extreme performance for creators and gamers	Excellent for gaming and everyday use	Power efficient hybrid performance
• 24 Cores (8 P + 16 E)	• 10 Cores (6 P + 4 E)	• 16 Cores	• 6 Cores	• 16 Cores (6 P + 8 E + 2 LP)
• 32 Threads	• 16 Threads	• 32 Threads	• 12 Threads	• 22 Threads
• Up to 6.0 GHz	• Up to 4.7 GHz	• Up to 5.7 GHz	• Up to 5.1 GHz	• Up to 4.8 GHz
• LGA 1700 Socket	• LGA 1700 Socket	• AM5 Socket	• AM5 Socket	• Intel 4 Process

MOBILE / LAPTOP CPUs For laptops, thin & light devices

Apple M3	AMD Ryzen 7 7840U	Intel Core i7-1360P	Qualcomm Snapdragon X Elite	Intel Core Ultra 5 125U
Powerful and efficient for MacBooks	Great performance with amazing battery life	High performance mobile processor	Arm-based chip for Windows laptops	Efficient performance for everyday use
• 8 CPU Cores	• 8 Cores	• 12 Cores (4 P + 8 E)	• 12 Cores	• 12 Cores (2 P + 8 E + 2 LP)
• 16 GPU Cores	• 16 Threads	• 16 Threads	• Up to 4.3 GHz	• 16 Threads
• 16-Core Neural Engine	• Up to 5.1 GHz	• Up to 5.0 GHz	• Adreno GPU	• Up to 4.3 GHz
• Unified Memory	• Radeon 780M Graphics	• Intel Iris Xe Graphics	• Very low power	• Intel 4 Process

WHAT IS A CPU?
The CPU (Central Processing Unit) is the main processor. It executes instructions, runs your operating system, apps, and handles general tasks.

COMMON CPU USES

- Operating System
- Web Browsing
- Productivity Apps
- Multitasking
- Programming
- Calculations

GPU vs CPU

Feature	CPU	GPU
Designed For	Generality	Parallel Processing
Cores	Few (4 - 24)	Many (hundreds - thousands)
Best At	General tasks	Graphics, AI, Compute
Used For	OS, Apps, Games Logic	Gaming, 3D, AI, Video Editing
Power Use	Lower	Higher
Examples	Intel Core, AMD Ryzen, Apple M	NVIDIA GeForce, RTX, AMD Radeon

GRAPHICS PROCESSING UNITS (GPUs)

CONSUMER / GAMING GPUs For gaming, content creation, and everyday users

NVIDIA GeForce RTX 4090	NVIDIA GeForce RTX 4070 Ti SUPER	NVIDIA GeForce RTX 4060	AMD Radeon RX 7900 XT	AMD Radeon RX 7600
The ultimate gaming GPU	High performance for 1440p / 4K gaming	Great for 1080p gaming and content creation	Great for 1080p gaming and streaming	Great value for 1080p gaming
• 36GB GDDR6X	• 16GB GDDR6X	• 8GB GDDR6	• 16GB GDDR6	• 8GB GDDR6
• 128 CUDA Cores	• 8448 CUDA Cores	• 3572 CUDA Cores	• 3792 Stream Processors	• 3792 Stream Processors
• Ray Tracing	• Ray Tracing	• Ray Tracing	• Ray Tracing	• Ray Tracing
• DLSS 3	• DLSS 3	• DLSS 3	• FSR 3	• FSR 3

PROFESSIONAL / WORKSTATION GPUs For professionals, engineers, creators, and AI development

NVIDIA RTX A6000 Ada	NVIDIA RTX A4500	NVIDIA RTX A2000	NVIDIA L40S	AMD Radeon PRO W7900
High-end professional graphics and compute	Great for 3D design, rendering, and AI	Compact and efficient workstation GPU	Powerful for AI training and large workloads	Professional performance for creators
• 48GB GDDR6	• 16GB GDDR6	• 12GB GDDR6	• 48GB GDDR6	• 48GB GDDR6
• 1074 CUDA Cores	• 4608 CUDA Cores	• 3328 CUDA Cores	• 18112 CUDA Cores	• 6144 Stream Processors
• ECC Memory	• ECC Memory	• ECC Memory	• Tensor Cores	• ECC Memory
• Optimized Drivers	• ECC Memory	• ECC Memory	• Data Center Ready	• Professional Drivers

WHAT IS A GPU?
The GPU (Graphics Processing Unit) specializes in processing thousands of tasks at once. It renders images, videos, and accelerates AI and scientific workloads.

COMMON GPU USES

- Gaming
- 3D Rendering
- Video Editing
- Live Streaming
- AI / Machine Learning
- Scientific Simulations

DID YOU KNOW?
GPUs have thousands of smaller cores designed to work on many tasks at the same time, which makes them perfect for graphics, AI, and high-performance computing.

Graphics and AI hardware — from rendering to neural networks

- The same GPU hardware runs games and neural networks.
- Tensor / matrix cores accelerate AI operations.
- Local AI models now run on consumer GPUs.

GRAPHICS & AI HARDWARE

The Key Processors Powering Modern Graphics, Computing & Artificial Intelligence

1. DEDICATED GPU

Graphics Processing Unit



A powerful, standalone chip designed primarily for high-performance graphics rendering and parallel computation.

BEST FOR

- Gaming
- 3D Rendering
- Video Editing
- AI/ML Training (High Performance)

KEY FEATURES

- Thousands of high-performance cores (CUDA cores)
- High memory bandwidth (GDDR6/6X)
- Designed for massive parallel workloads
- Typically 75W ~ 600W+ TDP

EXAMPLE PRODUCTS

NVIDIA GeForce RTX 4090, AMD Radeon RX 7900 XTX, Intel Arc A770

WHERE USED

Gaming PCs, Workstations, Content Creation, AI Training Servers

2. INTEGRATED GPU

Integrated Graphics



Built into the CPU or SoC and shares system memory. Optimized for efficiency and everyday computing.

BEST FOR

- Everyday Computing
- Web Browsing & Office Apps
- 4K Video Playback
- Light Gaming

KEY FEATURES

- Shares memory with system (RAM)
- Lower power consumption
- Smaller die size & cost-effective
- Good for general-purpose tasks

EXAMPLE PRODUCTS

Intel Iris Xe Graphics, AMD Radeon Graphics (GPU), Qualcomm Adreno Graphics

WHERE USED

Laptops, Ultrabooks, Desktops, Mobile Devices

3. AI ACCELERATOR (NPU)

Neural Processing Unit



Specialized processor designed specifically for AI/ML workloads with high efficiency at low power.

BEST FOR

- On-device AI (Inference)
- Voice & Image Recognition
- Real-time Translation
- Low-power AI Apps

KEY FEATURES

- Optimized for AI inference
- Extremely power efficient
- Built into SoCs (CPU + NPU)
- Handles neural networks efficiently

EXAMPLE PRODUCTS

Apple Neural Engine, Snapdragon Qualcomm Hexagon NPU, Microsoft Pluton / NPU

WHERE USED

Smartphones, Laptops, Tablets, Edge Devices, IoT

4. AI ACCELERATOR CARD (ENTERPRISE)

High-Performance AI Card



Add-in cards designed for data centers and enterprises to accelerate AI training and inference at scale.

BEST FOR

- AI Model Training (Large Scale)
- High-Performance Inference
- Data Analytics
- HPC & Scientific Computing

KEY FEATURES

- Massive compute power & memory (HBM2e/HBM3)
- High bandwidth (NVLink / PCIe)
- Optimized for datacenter workloads
- Scalable across multiple cards

EXAMPLE PRODUCTS

NVIDIA H100 Tensor Core GPU, AMD Instinct MI300X, Intel Gaudi 3 AI Accelerator

WHERE USED

Data Centers, AI Labs, Cloud Platforms, Enterprise AI Systems

5. TPU (TENSOR PROCESSING UNIT)

Google's AI Accelerator



Custom-built by Google for machine learning workloads, optimized for TensorFlow and large-scale AI training.

BEST FOR

- AI/ML Training (Google Cloud)
- Deep Learning Workloads
- Large-Scale Inference
- Data Center AI at Scale

KEY FEATURES

- Designed specifically for tensor operations
- Very high throughput
- Optimized for TensorFlow
- High energy efficiency

EXAMPLE PRODUCTS

TPU v4, TPU v5e, TPU v5e

WHERE USED

Google Cloud, AI Research, Machine Learning Infrastructure

QUICK COMPARISON		Graphics & Parallel Compute	General Purpose Efficiency	AI Inference (Edge, Low Power)	AI Training & Inference (Data Center)	AI Training (Google Cloud)
Power Usage		High	Low	Very Low	High	High (Optimized)
Typical Location		Discrete Card (PCIe)	Built into CPU/SoC	Built into SoC	Add-in Card (PCIe)	Custom Board (Data Center)

i Different hardware for different needs — from gaming to AI at scale!

Note: Specifications and examples are representative and may change over time.

Modern AI PC hardware — CPU + GPU + NPU together

- NPU (Neural Processing Unit) is a dedicated AI chip.
- Copilot+ PCs use CPU, GPU and NPU together.
- NPU is power-efficient — great on laptops.

MODERN AI PC HARDWARE

Powering the next generation of intelligent, faster, and more efficient experiences.

1 NEURAL PROCESSING UNIT (NPU)

WHAT IT IS
A dedicated processor built specifically for AI workloads.

WHY IT MATTERS

- Runs AI tasks locally and efficiently
- Improves battery life
- Powers real-time features (e.g., Recall, Live Captions, Studio Effects)

AI Acceleration On-Device AI Better Battery Life

2 COPILOT+ PC ARCHITECTURE

WHAT IT IS
A new class of Windows PCs designed for AI, built around the NPU, powerful CPU, GPU, and fast memory.

WHY IT MATTERS

- Optimized for AI experiences
- Balanced performance and efficiency
- Unlocks exclusive Copilot+ PC features
- Smarter by Design

Optimized Stack AI-Ready Smarter by Design

3 DDR5 MEMORY

WHAT IT IS
The latest high-speed memory for desktops and laptops.

WHY IT MATTERS

- Higher bandwidth
- Better multitasking
- Supports demanding AI and creative apps

Faster Performance Greater Capacity Improved Efficiency

4 LPDDR5X MEMORY

WHAT IT IS
Low-power memory designed for thin, light devices.

WHY IT MATTERS

- Ultra-fast speeds
- Lower power usage
- Longer battery life in thinner designs

Low Power High Speed Longer Battery Life

5 PCIe 5.0 SSD

WHAT IT IS
Next-gen storage with blazing fast read/write speeds.

WHY IT MATTERS

- Faster boot and load times
- Handles large AI models and datasets
- Future-proof bandwidth

Extreme Speed High Capacity Reliable Performance

6 AI GPU

WHAT IT IS
Graphics processors optimized for AI and parallel workloads.

WHY IT MATTERS

- Accelerates AI training and inference
- Great for content creation and 3D rendering
- Enables local AI apps

AI Acceleration Content Creation 3D & Simulation

7 WI-FI 7 MODULE

WHAT IT IS
The latest Wi-Fi standard offering ultra-fast, low-latency wireless connectivity.

WHY IT MATTERS

- Multi-gigabit speeds
- Lower latency
- Better in crowded environments

Ultra-Fast Low Latency More Reliable

8 USB4 / THUNDERBOLT 5 CONTROLLER

WHAT IT IS
Next-gen connectivity controller for super-fast data, video, and power delivery.

WHY IT MATTERS

- Up to 80 Gbps bandwidth
- Connect high-res displays and docks
- Charge devices and transfer data simultaneously

80 Gbps Display & Docking Power Delivery

BUILT FOR AI. READY FOR YOU.

Modern AI PCs combine specialized hardware and smarter architecture to deliver powerful on-device AI experiences—securely, efficiently, and intelligently.

WHAT CAN YOU DO?

Real-time translation AI image generation Video enhancement Code completion Smart collaboration Personalized experiences

THE AI PC ADVANTAGE

- ✓ Run AI locally, even without the cloud
- ✓ More responsive and seamless experiences
- ✓ Better privacy and data security
- ✓ Optimized power for all-day productivity

★ Smarter Hardware. Smarter Experiences. The foundation of the AI-powered future. Choose an AI PC built for what's next.

Modern AI PC hardware — CPU + GPU + NPU together

CPU cooler — traditional air cooler

- Heatsink + fan moves heat from CPU to case air.
- Reliable, cheap, no leaks.
- Thermal paste sits between CPU and heatsink.

CPU COOLER (AIR COOLER)

Keeping Your CPU Cool for Peak Performance

A CPU air cooler dissipates heat from the processor using a heatsink and fan. It draws heat away from the CPU and transfers it into the air, keeping temperatures low and performance high.

HOW IT WORKS

- 1 The CPU generates heat when in use.
- 2 The heat is transferred to the cooler's baseplate.
- 3 Heat travels up through the heat pipes into the aluminum fins.
- 4 The fan pushes cool air through the fins, carrying heat away.
- 5 Warm air is expelled into the case and exhausted by case fans.

AIRFLOW INSIDE THE CASE

Cool Air In → Front Intake Fans bring cool air into the case.

Warm Air Out → Rear and top exhaust fans remove warm air from the case.

AIR COOLER COMPONENTS

- Fan:** Moves air through the heatsink.
- Aluminum Fins:** Large surface area dissipates heat into the air.
- Heat Pipes:** Conduct heat from the base to the fins efficiently.
- Baseplate:** Directly contacts the CPU and absorbs heat.
- Mounting System:** Secures the cooler to the motherboard.

TYPES OF AIR COOLERS

- TOWER COOLER:** Most common type. Offers excellent cooling performance and fits most PC cases.
- TOP-FLOW COOLER:** Blows air downward onto the motherboard. Ideal for small cases or tight spaces.

KEY FEATURES

- EFFICIENT COOLING:** Heat pipes and fins work together to dissipate heat effectively.
- QUIET OPERATION:** Large fans can move more air at lower speeds, reducing noise.
- EASY INSTALLATION:** Simple mounting systems support all major CPU sockets.
- RELIABLE & DURABLE:** No moving parts other than the fan, making it long-lasting.
- COST EFFECTIVE:** Excellent performance at an affordable price compared to liquid coolers.
- NO MAINTENANCE:** Just install and forget. No refilling or special maintenance required.

TIPS FOR BEST PERFORMANCE

- ✓ Apply a high-quality thermal paste.
- ✓ Ensure good airflow in your case.
- ✓ Keep the cooler and case fans clean.
- ✓ Choose the right cooler size for your CPU (TDP).
- ✓ Monitor temperatures for optimal performance.

DID YOU KNOW?

A good air cooler can keep high-performance CPUs running cool and stable without the cost and complexity of liquid cooling.

COMPATIBILITY

SOCKET SUPPORT (Examples)

INTEL	AMD
LGA 1851 / 1700	AM5 / AM4
LGA 1200 / 1151	AM3+ / AM3
LGA 1150 / 1155	AM2+ / AM2
LGA 1156	FM2+ / FM2

CLEARANCE CHECK

Check 84mm height for clearance with the cooler.

Ensure the cooler height fits your case.

EXAMPLE AIR COOLER SPECIFICATIONS (Typical Tower Cooler)

DIMENSIONS	WEIGHT	FAN SIZE	FAN SPEED	AIRFLOW	NOISE LEVEL	TDP SUPPORT
152 x 76 x 154 mm (L x W x H)	650 g	120 mm	400 – 1500 RPM (±10%)	59 CFM (Max)	25.6 dBA (Max)	Up to 180W

Liquid CPU cooler — AIO (All-In-One)

- Sealed pump moves coolant to a radiator.
- Better for very hot CPUs (i9/Ryzen 9).
- Pump can fail — watch for temperature spikes.

LIQUID CPU COOLER (AIO)

Advanced Cooling for Maximum Performance

An All-In-One (AIO) liquid cooler uses liquid to transfer heat away from the CPU, then dissipates it through a radiator. It provides superior cooling compared to air coolers, ideal for high-performance CPUs and overclocking.

AIO COMPONENTS

- RADIATOR** Aluminum radiator contains thin fins that maximize surface area for heat dissipation.
- FANS** Mounted on the radiator to push air through the fins and remove heat.
- TUBES** Durable, low-permeation tubes carry coolant between the pump and radiator.
- PUMP** Circulates coolant through the system. Usually located in the pump housing.
- COLD PLATE** Copper plate makes direct contact with the CPU to absorb heat efficiently.

HOW IT WORKS

- The CPU generates heat during operation.
- Coolant in the cold plate absorbs the heat.
- The pump moves the heated coolant through the tubes to the radiator.
- The radiator's fins and fans dissipate the heat into the air.
- The cooled liquid returns to the pump to repeat the cycle.

KEY BENEFITS

- SUPERIOR COOLING** More efficient heat transfer than air coolers, lower temperatures.
- QUIET OPERATION** Fans run at lower speeds for the same cooling performance.
- BETTER FOR OVERCLOCKING** Keeps high-performance CPUs cool under heavy loads.
- SLEEK & MODERN LOOK** Clean installation with no bulky heatsink on the CPU.
- SPACE SAVING** Frees up space around the CPU socket for better airflow.

AIO vs AIR COOLER

AIO LIQUID COOLER	AIR COOLER
✓ Better cooling performance	✓ Good cooling performance
✓ Handles high TDP CPUs	✗ May struggle at high TDP
✓ Better for overclocking	✗ Limited overclocking headroom
✓ Quieter at high loads	✗ Can be noisier under load
✓ Clean and sleek look	✗ Bulky and taller
✓ Higher cost	✓ More affordable

COMMON RADIATOR SIZES

Size	Fans	Benefits
120mm	(1 Fan)	• Good for low to mid-range CPUs • Fits in most small cases • Best for standard builds
240mm	(2 Fans)	• Great balance of performance & size • Most popular size • Ideal for gaming PCs
360mm	(3 Fans)	• Best cooling performance • Ideal for high-end CPUs • Requires larger case support

COLD PLATE CLOSE-UP

- Copper Base** High thermal conductivity for efficient heat transfer.
- Micro Fins** Increase surface area to absorb more heat from the CPU.
- Pre-Applied Thermal Paste** Promotes optimal contact and easy installation.

INSTALLATION OVERVIEW

- Prepare** Clean CPU surface and apply thermal paste (if not pre-applied).
- Mount Brackets** Install the mounting brackets to the motherboard.
- Mount Pump Block** Secure the pump block onto the CPU using the screws.
- Mount Radiator** Install the radiator in the case and secure it.
- Connect Cables** Connect PWM and RGB cables to the motherboard and power.

MAINTENANCE TIPS

- Ensure radiator fans and case fans are clean.
- Check for air bubbles or unusual pump noise.
- Keep your case well-ventilated.
- AIO coolers are maintenance-free, but replace if performance drops or leaks occur.

DID YOU KNOW?

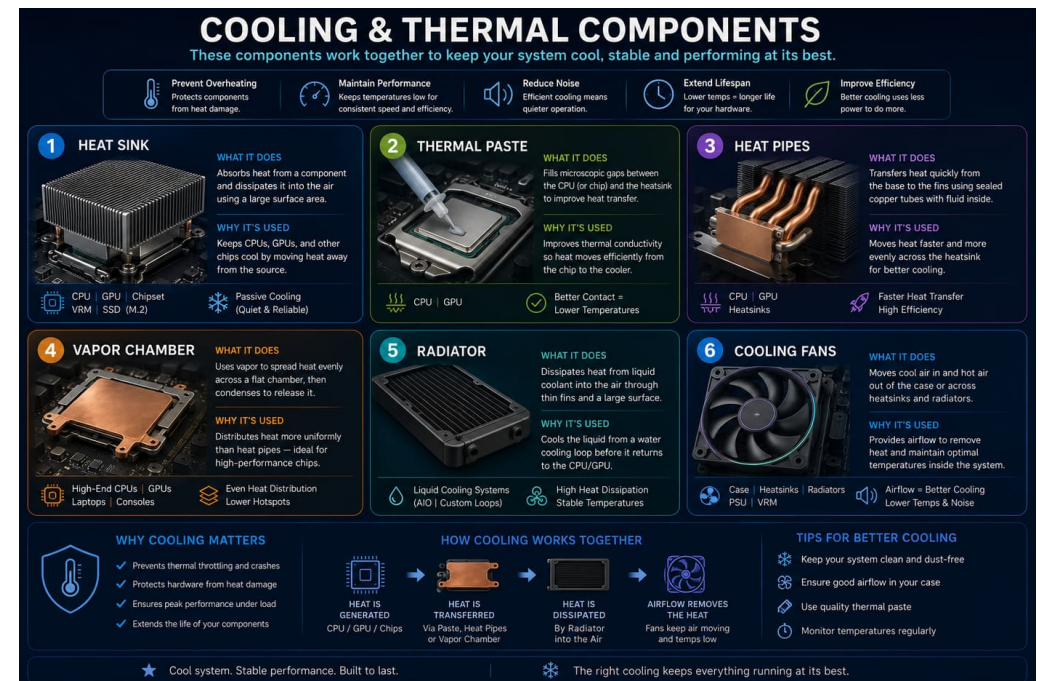
A liquid cooler can reduce CPU temperatures by 10–20°C compared to high-end air coolers, depending on the CPU and workload.

SPECIFICATIONS (EXAMPLE 240mm AIO)

Radiator Material	Aluminum
Radiator Size	277 x 120 x 27 mm
Fan Size	120 x 120 x 25 mm
Fan Speed	600 – 1800 RPM (±10%)
Airflow	21 – 78 CFM
Noise Level	18 – 30 dBA
Pump Speed	2000 – 3000 RPM (±10%)
Socket Support	Intel LGA 1851/1700/1200/1155 AND: AM5 / AM4
Warranty	3 – 6 Years

Cooling and thermal components — the whole thermal system

- Case fans, CPU cooler, VRM & M.2 heatsinks, paste.
- Airflow direction: in at the front, out at the rear.
- Dust is enemy #1 — clean filters regularly.



Memory (RAM) — how the modules work

- Volatile — loses everything on power off.
- More RAM = more apps open at once without slowing down.
- DDR4 and DDR5 are the current standards.



DIMM slots — where RAM plugs in

- Colour-coded pairs enable dual-channel mode.
- Populate matching slots for double the bandwidth.
- Check maximum supported speed in the manual.

DIMM SLOTS (DUAL IN-LINE MEMORY MODULE)

THE MEMORY CONNECTORS ON YOUR MOTHERBOARD

DIMM slots are the long connectors on a motherboard that hold the RAM modules securely in place. They allow your system to access high-speed memory for faster multitasking, applications, and performance.

KEY FUNCTIONS

- Secure Connection**
Holds RAM modules firmly in place.
- High-Speed Data Transfer**
Enables fast communication between RAM and the CPU.
- Supports Multiple Configurations**
Single or dual channel for better performance.
- Determines Capacity & Speed**
Supports the maximum memory capacity and speed supported by the board.

DIMM SLOTS ON A MOTHERBOARD (Typical)

DIMM Slots (4 slots shown)
These are the connectors where RAM modules are installed.

Slot Key (Notch)
The notch in the RAM module must match the key in the slot for correct installation.

Retention Clips
Clips on both ends lock the RAM module in place. Push outwards to release the module.

Channel Color Coding
Slots are color-coded (e.g., Blue/Black) to indicate memory channels for optimal performance.

DIMM KEY (NOTCH) – ENSURES CORRECT FIT

Correct Fit
The notch on the RAM module aligns with the key in the slot.

Incorrect Fit
If the notch does not align, the RAM module will not fit.

MEMORY CHANNELS EXPLAINED

Single Channel
RAM modules installed in one channel only.

Dual Channel (Recommended)
Modules installed in matching slots of both channels.

Quad Channel (High-End)
Requires compatible CPU and motherboard.

Channel A (Blue) Channel B (Black)

TYPES OF DIMM (BY MEMORY GENERATION)

DDR3 DIMM	DDR4 DIMM	DDR5 DIMM
240-Pin	288-Pin	288-Pin
Voltage: 1.5V	Voltage: 1.2V	Voltage: 1.1V
Use: Older Systems	Use: Most Common	Use: Latest Systems

HOW TO INSTALL A DIMM

- Open the retention clips on both ends of the slot.
- Align the notch on the RAM module with the key in the slot.
- Firmly press down on the module evenly until the clips snap into place.
- Ensure both clips are locked and the module is secure.

Always handle RAM by the edges and avoid touching the gold contacts.

TIPS & BEST PRACTICES

- Use the RAM type and speed recommended by your motherboard.
- Populate slots as per the motherboard manual for best performance.
- For dual channel, install matching modules in the correct slots (e.g., A2 and B2).
- Mixing different RAM sizes or speeds may run at the lowest common speed.
- Ensure DIMMs are fully seated and clips are locked.
- Keep DIMM slots and modules clean and free from dust.

DID YOU KNOW?

Using dual channel memory can increase memory bandwidth by up to 100%, significantly improving system performance in many tasks and applications.

QUICK REFERENCE – HOW MANY RAM MODULES?

Slots Available	1 Slot	2 Slots	4 Slots
Best Practice	1 DIMM	2 DIMMs (Dual Channel)	2 or 4 DIMMs (Dual / Quad Channel)

COMMON DIMM SLOT ISSUES

- System not posting: RAM may not be fully seated.
- Lower performance: RAM not in the correct slots for the channel.
- Beep codes or errors: Incompatible or faulty RAM.

PRO TIP

Always refer to your motherboard manual for the exact DIMM slot population guide and limitations.

Storage hardware — the full family (HDD, SSD, NVMe)

- HDD: spinning platters, cheap and large, slow.
- SATA SSD: no moving parts, fast enough for most users.
- NVMe SSD: plugs into M.2, blazing fast.

STORAGE HARDWARE

Different types of storage devices for every need

Storage hardware is used to store, access, and protect data. From ultra-fast NVMe SSDs to high-capacity HDDs and NAS drives, each type is designed for specific use cases.

KEY INTERFACES

NVMe / PCIe Ultra-fast | SATA Widely compatible

1 NVMe SSD (M.2 PCIe)



Uses PCIe interface via M.2 slot for the fastest performance.

- Blazing fast read/write speeds
- Low latency
- Ideal for demanding applications and gaming

Best for: High-end PCs, Gaming, Video Editing, Workstations

Interface: PCIe 3.0 / 4.0 / 5.0 (x4)

Form Factor: M.2 2280 / 2230

2 SATA SSD (2.5-inch)



Uses SATA interface for broad compatibility.

- Much faster than HDDs
- Reliable and durable
- Great for everyday computing upgrades

Best for: Laptops, Desktops, Everyday Use, Upgrades

Interface: SATA III (6 Gb/s)

Form Factor: 2.5-inch (7mm)

3 EXTERNAL SSD (Portable)



Portable SSD with USB connection for on-the-go storage.

- Fast transfer via USB 3.2 / USB-C
- Rugged and compact
- Ideal for backups and travel

Best for: Travel, Backups, Media Files, On-the-Go

Interface: USB 3.2 Gen 2 / USB-C

Form Factor: Portable (Pocket Size)

4 ENTERPRISE SSD (Data Center)



Built for 24/7 workloads in enterprise and data center environments.

- High endurance (DWPD)
- Consistent performance
- Power loss protection & advanced reliability

Best for: Servers, Virtualization, Databases, Data Centers

Interface: SATA / SAS / PCIe (U.2, U.3, M.2)

Form Factor: 2.5-inch / U.2 / U.3 / M.2

5 NAS DRIVE (Network Attached Storage)



Designed for NAS systems and always-on network environments.

- Optimized for 24/7 operation
- High reliability and vibration resistant
- Supports RAID and multi-user access

Best for: Home NAS, Small Business, Media Server, File Sharing

Interface: SATA III (6 Gb/s)

Form Factor: 3.5-inch

6 HDD (Hard Disk Drive)



Traditional spinning disk drive for high-capacity storage.

- Large storage capacity
- Lowest cost per TB
- Ideal for archives and backups

Best for: Archiving, Backups, Surveillance, Bulk Storage

Interface: SATA III (6 Gb/s)

Form Factor: 3.5-inch / 2.5-inch

COMPARISON AT A GLANCE

Type	Interface	Speed	Capacity	Durability	Best Use
NVMe SSD	PCIe (NVMe)	***** (Up to 14 GB/s)	Up to 8 TB	High	High performance, gaming, content creation
SATA SSD	SATA III (6 Gb/s)	***** (Up to 560 MB/s)	Up to 4 TB	High	Daily use, upgrades
External SSD	USB 3.2 / USB-C	***** (Up to 2000 MB/s)	Up to 8 TB	High	Portability, backups, media transfer
Enterprise SSD	SATA / SAS / PCIe	***** (Consistent)	Up to 15 TB	Very High	Servers, databases, data centers
NAS Drive	SATA II	***** (Up to 350 MB/s)	Up to 20 TB*	High	NAS, always-on systems, file sharing
HDD	SATA II	***** (Up to 200 MB/s)	Up to 24 TB*	Medium	Archiving, backups, bulk storage

HOW THEY CONNECT



M.2 NVMe SSD connects to M.2 slot on motherboard.

SATA SSD / HDD / NAS connects via SATA Cable to SATA Port.

External SSD connects via USB-C / USB-A Cable to USB Port.

Enterprise SSD connects via U.2 / U.3 / SATA / SAS to Server / Rackmount.

TIPS

- Use NVMe SSDs for the fastest system and application performance.
- Use SATA SSDs for a great balance of speed and price.
- Back up important data regularly to an external drive or NAS.
- Choose enterprise or NAS drives for 24/7 reliability.
- HDDs are best for large, infrequently accessed data.

DID YOU KNOW?

A 1TB HDD can hold about 250 movies, while a 1TB NVMe SSD can transfer those same movies in just a few seconds!

SSD (NVMe M.2) — the modern fast SSD

- Talks directly over PCIe — no SATA bottleneck.
- Up to 10× faster than SATA SSDs.
- Runs hot — usually needs a small heatsink.

SSD (NVMe M.2)

Blazing Fast Storage for Modern Computers

An NVMe M.2 SSD is a high-performance solid-state drive that uses the NVMe (Non-Volatile Memory Express) protocol and connects directly to the motherboard via the M.2 slot for ultra-fast data transfer and low latency.

WHAT IS AN SSD?

SSD stands for Solid State Drive. Unlike traditional hard drives (HDDs) that use moving parts, SSDs use flash memory chips to store data. This makes them faster, quieter, more reliable, and more energy efficient.

NVMe M.2 SSD OVERVIEW

NAND Flash Memory Stores your data (persistent storage)

DRAM Cache (Optional) Speeds up performance by caching data

Controller Manages data transfer between NAND and the system

V-NAND SSD 970 EVO Plus NVMe M.2 1TB

M Key Fits into M.2 slot (It's fixed on the motherboard)

NVMe Interface Uses PCIe lanes for ultra-fast data transfer

HOW NVMe M.2 SSD WORKS

Data flows directly between the CPU and the NVMe SSD over PCIe lanes, eliminating the slower SATA interface and reducing latency.

KEY BENEFITS OF NVMe M.2 SSD

- ULTRA FAST SPEEDS** Up to 7000+ MB/s read speeds — much faster than SATA SSDs and HDDs.
- LOW LATENCY** NVMe protocol is optimized for speed and responsiveness.
- HIGH RELIABILITY** No moving parts means less wear and tear and better durability.
- ENERGY EFFICIENT** Consumes less power and runs cooler than traditional drives.

NVMe vs SATA: WHAT'S THE DIFFERENCE?

SATA SSD	NVMe M.2 SSD
Uses SATA interface	Uses PCIe interface
Max speed: ~600 MB/s	Max speed: 3,000 – 7,000+ MB/s
Uses SATA cable	Connects directly to M.2 slot
AHCI protocol	NVMe protocol
Higher latency	Lower latency
Older technology	Modern & future-ready

NVMe PROTOCOL ADVANTAGE

NVMe is designed specifically for SSDs and allows thousands of commands to be processed in parallel.

AHCI (SATA) Processes 1 command at a time

NVMe (PCIe) Processes thousands of commands at once

Lower Latency | Higher Throughput | Better Efficiency

KEY SPECIFICATIONS

Interface	PCIe (Gen3 x4 / Gen4 x4 / Gen5 x4)
Protocol	NVMe (Non-Volatile Memory Express)
Form Factor	M.2 (2280, 2240, 2260, etc.)
Capacities	250GB, 500GB, 1TB, 2TB, 4TB, 8TB+
Sequential Read Speed	Up to 7,000+ MB/s (Gen4)
Sequential Write Speed	Up to 6,500+ MB/s (Gen4)
Endurance (TBW)	150 TBW – 6800+ TBW (varies by model)
MTTF (Reliability)	Up to 1.5 Million Hours

M.2 KEY TYPES

M.2 connectors come in different "keys" to prevent incorrect insertion.

- M KEY** 5 pins wide. Used for NVMe (PCIe) SSDs.
- B KEY** 6 pins wide. Used for SATA SSDs (or PCIe x2).
- B+M KEY** Both B and M. Supports SATA or NVMe (PCIe) SSDs.

M.2 SSD FORM FACTORS (BY LENGTH)

All are 22mm wide, but lengths vary.

- 2230** (22mm x 30mm): Common in compact devices and form factors.
- 2242** (22mm x 42mm): Tablets and ultrabooks.
- 2260** (22mm x 60mm): Some laptops and small PCs.
- 2280** (22mm x 80mm): Most common for desktops and laptops.

DID YOU KNOW?

An NVMe M.2 SSD can be up to 12x faster than a SATA SSD and up to 100x faster than a traditional HDD!

M.2 slots — where NVMe SSDs live

- Long, thin slot on the motherboard.
- Uses PCIe lanes for high speed.
- Watch for shared bandwidth with SATA on some boards.

M.2 SLOTS

HIGH-PERFORMANCE STORAGE IN A SMALL FORM FACTOR

M.2 slots allow you to install M.2 SSDs that provide ultra-fast storage performance in a compact form factor. They connect directly to the motherboard and use PCIe lanes (or SATA) for data transfer.

KEY FUNCTIONS

- High Speed**
Supports PCIe NVMe SSDs with exceptional read/write speeds.
- Compact Design**
M.2 SSDs are much smaller than traditional 2.5" drives.
- Versatility**
Supports different lengths and both PCIe and SATA interfaces.
- Direct Connection**
Connects directly to the motherboard for lower latency and better performance.

M.2 SLOTS ON A MOTHERBOARD (EXAMPLES)

Not all motherboards have the same number of M.2 slots or support all M.2 types.

PARTS OF AN M.2 SLOT

- M.2 Connector**
The slot where the M.2 SSD is inserted.
- Mounting Hole / Standoff**
Used to secure the SSD at different lengths.
- Screw**
Keeps the M.2 SSD firmly in place.
- Heatsink (Optional)**
Helps maintain optimal temperatures for better performance.

M.2 KEYING (NOTCHES)

The notch on the connector determines compatibility.

- Key M (Socket 3)**
For NVMe (PCIe) & SATA SSDs
- Key B (Socket 2)**
For SATA SSDs
- Key E (Socket 1)**
For Wi-Fi / Bluetooth and some PCIe devices

M.2 SIZES (LENGTH)

The motherboard will list which sizes are supported for each M.2 slot.

M.2 INTERFACES

PCIe (NVMe)

Uses PCI Express lanes for maximum performance.
Common: PCIe 3.0 x4 / PCIe 4.0 x4 / PCIe 5.0 x4

SATA

Uses SATA bus for data transfer.
Typical speed: Up to 600 MB/s

TIPS & BEST PRACTICES

- Use an M.2 SSD in the highest bandwidth slot for best performance.
- Ensure the M.2 SSD is compatible (key, interface, and size).
- Use a heatsink if your motherboard does not include one.
- Keep your BIOS/UEFI updated for maximum compatibility and performance.
- Do not overtighten the screw.

HOW TO INSTALL AN M.2 SSD

- Locate the M.2 Slot**
Find the M.2 slot on your motherboard.
- Remove Screw**
Remove the screw from the mounting hole (standoff).
- Insert SSD**
Insert the M.2 SSD into the slot at a 30° angle.
- Press Down**
Gently press the SSD down flat so the hole aligns with the standoff.
- Secure with Screw**
Secure the SSD with the screw you removed.

Handle M.2 SSDs by the edges and avoid touching the gold contacts.

M.2 LANE SHARING

Some M.2 slots share bandwidth with other ports. Using certain M.2 slots may disable SATA ports or PCIe slots.

Example:

- Using M.2_1 (PCIe 4.0 x4) disables SATA_5 and SATA_6.

Always check your motherboard manual for lane sharing details.

DID YOU KNOW?

M.2 SSDs can be up to 7x faster than SATA SSDs in real-world performance (NVMe PCIe 4.0 vs SATA).

INTERFACE SPEED COMPARISON (THEORETICAL MAXIMUM)

Interface	PCIe 3.0 x4 / SATA III	PCIe 4.0 x4	PCIe 5.0 x4
Bandwidth	~3.94 GB/s	~7.88 GB/s	~15.75 GB/s
Typical SSD Speed	Up to 560 MB/s	Up to 7,000 MB/s	Up to 14,000+ MB/s

COMMON M.2 SLOT ISSUES

- SSD not detected:** Check seating, compatibility, and BIOS settings.
- Slow speeds:** Ensure the SSD is in a PCIe slot, not a SATA M.2 slot.
- Overheating:** Add or improve heatsink / airflow.
- Lane sharing:** Other ports may be disabled when using certain slots.

PRO TIP

For best performance, install your OS and most used applications on an M.2 NVMe SSD.

SATA ports — connecting SATA drives and optical drives

- One data cable + one power cable per drive.
- Used for SATA SSDs, HDDs and DVD drives.
- Slower than NVMe but very flexible.

SATA PORTS

CONNECTING STORAGE DEVICES

SATA (Serial ATA) ports on the motherboard allow you to connect 2.5"/3.5" hard drives, SSDs, and optical drives for data storage and transfer.

KEY FUNCTIONS

- Data Connection**
Provides a data interface between the motherboard and SATA devices.
- Power Efficient**
SATA is more power efficient compared to older IDE (PATA).
- High Speed**
Supports up to 6 Gb/s (SATA III) for fast data transfer.
- Hot Swappable (AHCI)**
Allows drives to be connected/disconnected while the system is running (with AHCI enabled).

SATA PORTS ON A MOTHERBOARD

① The number and arrangement of SATA ports may vary by motherboard model.

SATA PORT TYPES (BY SPEED)

SATA Version	Max Speed	Color (Common)	Backward Compatibility
SATA I	1.5 Gb/s	Gray	Yes
SATA II	3 Gb/s	Black	Yes
SATA III	6 Gb/s	Blue	Yes

① SATA III ports are backward compatible with SATA II and SATA I devices.

SATA CONNECTORS (ON MOTHERBOARD)

L-shaped key ensures correct connection.

Provides power to the SATA device.

SATA DEVICES YOU CAN CONNECT

HOW SATA WORKS

① SATA uses two connections:
1. Data Cable (7-pin) → 2. Power Cable (15-pin)

HOW TO CONNECT A SATA DEVICE

- Connect SATA Data Cable**
Attach one end of the data cable to the motherboard SATA port.
- Connect to the Device**
Attach the other end of the data cable to the SATA port on the device.
- Connect Power Cable**
Connect the 15-pin power connector from the PSU to the device.
- Secure Cables**
Ensure both connectors are firmly seated.
- Power On & Detect**
Power on the system. The device should be detected in BIOS/UEFI.

SATA Information
SATA Port 1: Samsung SSD 870 EVO
SATA Port 2: WD WD1002FLEX-00HW0
SATA Port 3: KPMI H4050A

⚠ Always use the correct cables and ensure they are fully seated to avoid connection issues.

SATA CABLES

- Thin
- L-shaped
- Transfers data

- Wider
- Flat connector
- Provides power

TIPS & BEST PRACTICES

- Use SATA III (6 Gb/s) ports and cables for best performance.
- Enable AHCI mode in BIOS for maximum speed and features.
- Keep cables tidy and away from fans and heat sources.
- Check your motherboard manual for port limitations (e.g., some ports may be disabled when using M.2 SATA).
- For SSDs, place them on the primary SATA ports (often SATA1/2).

DID YOU KNOW?

SATA II (6 Gb/s) is 4x faster than SATA I (3 Gb/s) and 6x faster than SATA I (1.5 Gb/s).

SATA GENERATION SPEED COMPARISON

Generation	SATA I	SATA II	SATA III
Max Speed	1.5 Gb/s	3 Gb/s	6 Gb/s
Real-World Speed (approx.)	~150 MB/s	~300 MB/s	~600 MB/s

COMMON SATA PORT ISSUES

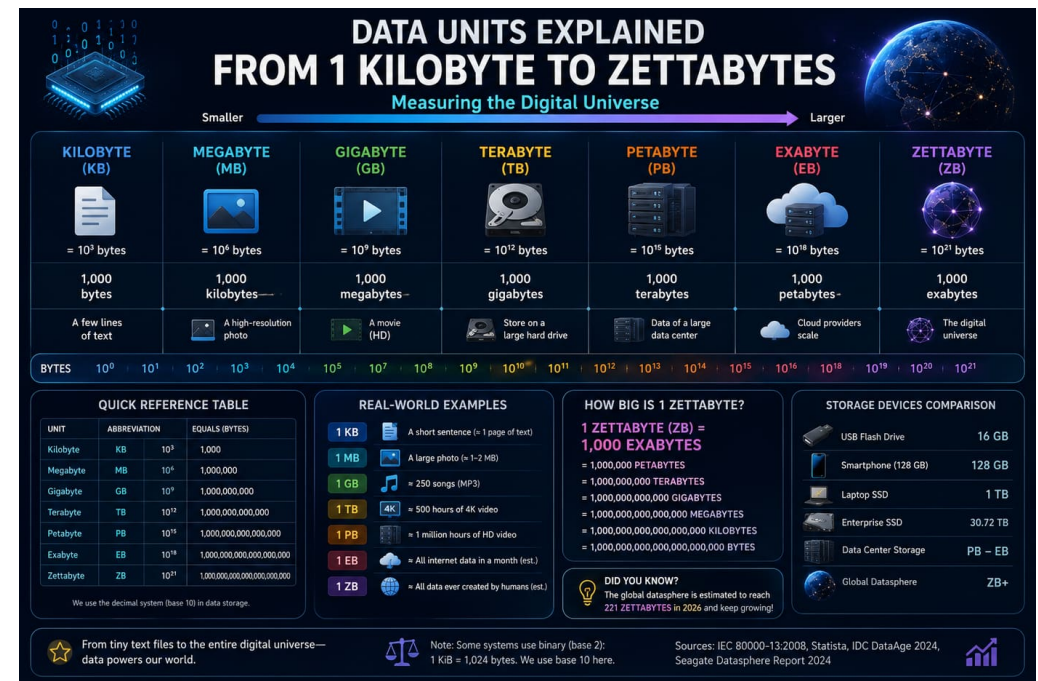
- Device not detected: Check cables and power connection.
- Slow performance: Ensure AHCI mode is enabled.
- Port not working: Try another SATA port or cable.
- M.2 SATA in use: Some SATA ports may be disabled.

PRO TIP

For the best performance, use an SSD with SATA II or a SATA III (6 Gb/s) port and enable AHCI in BIOS/UEFI.

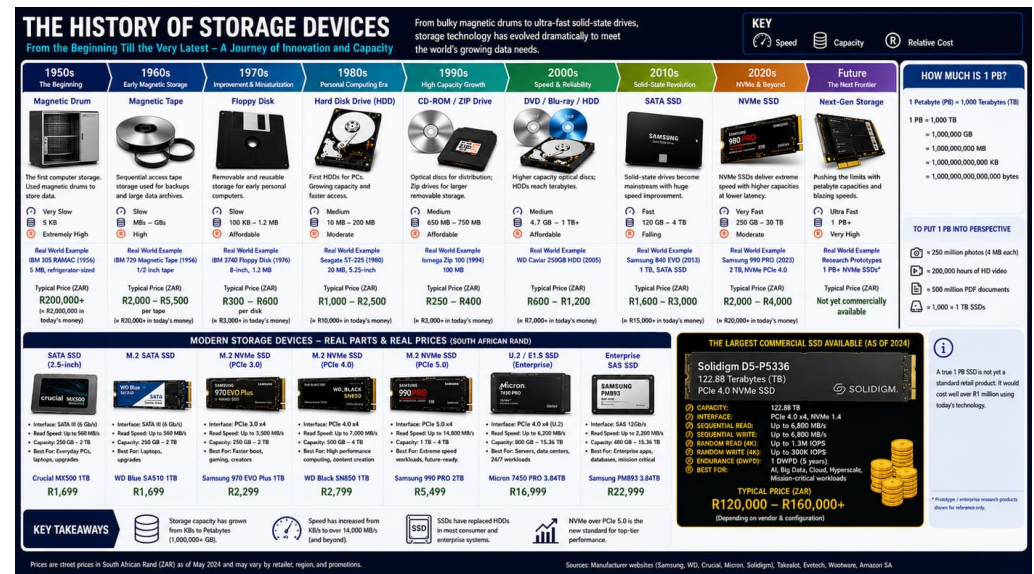
Data units explained — from 1 kilobyte to zettabytes

- KB !' MB !' GB !' TB !' PB !' EB !' ZB.
- Each step is roughly $\times 1000$.
- Users think in files; support thinks in units.



The history of storage devices — from magnetic drums to NVMe

- Punched cards !' tape !' drums !' HDD !' SSD !' NVMe.
- Storage got smaller, faster and cheaper every decade.
- History explains today's design choices.



How much information do we have in the world?

- Global data doubles every couple of years.
- Most new data is video, images and telemetry.
- Storage capacity is a business problem, not just tech.



How much information do we have in the world?

Power Supply Unit (PSU) — anatomy and connectors

- Converts AC mains to DC rails for every component.
- Rated in watts — must exceed system requirements.
- 80 PLUS rating = efficiency (Bronze < Gold < Titanium).

POWER SUPPLY UNIT (PSU)

The Heart That Powers Your PC

The Power Supply Unit (PSU) converts electricity from your wall outlet into the clean, regulated DC power that your computer's components need to run.

WHAT DOES A PSU DO?
It converts high-voltage AC (Alternating Current) from your wall outlet into low-voltage DC (Direct Current) and supplies it to all the components inside your computer safely and efficiently.

HOW A PSU WORKS

1. AC power enters the PSU from the wall outlet.
2. The EMI filter removes electrical noise.
3. The rectifier converts AC to high-voltage DC.
4. The inverter switches the DC at high frequency.
5. The transformer steps the voltage down.
6. The rectifier & filters produce stable DC outputs.
7. The PSU regulates the voltages and sends power to your PC components.

AC INPUT
Connects to your wall outlet using the included power cable.

POWER SWITCH
Turns the PSU on or off.

VOLTAGE SWITCH (If Available)
Selects input voltage (115V / 230V) based on your region.

120mm FAN
Cooling fan draws air in and pushes hot air out to keep the PSU cool.

KEY FUNCTIONS

- Converts AC to DC power
- Regulates voltage for stable performance
- Protects components from power issues (OCP, OVP, SCP, etc.)
- Keeps the system cool with built-in fans

EFFICIENCY RATINGS (80 PLUS)
PSUs are rated for energy efficiency. Higher efficiency means less wasted energy, less heat, and lower electricity costs.

80 PLUS STANDARD	80 PLUS BRONZE	80 PLUS SILVER	80 PLUS GOLD	80 PLUS PLATINUM	80 PLUS TITANIUM
Standard 80%	Bronze 82%	Silver 85%	Gold 87%	Platinum 90%	Titanium 92%

Typical Efficiency at 50% Load

DURABLE BUILD
Built with high-quality components for reliable and long-term use.

CABLES & CONNECTORS
Deliver power to all components through specific connectors.

CONNECTOR	USED FOR	IMAGE
24-Pin ATX	Motherboard	
4+4 Pin EPS (8-Pin CPU)	CPU Power	
6+2 Pin PCIe (8-Pin)	Graphics Cards	
SATA Power	SSDs, HDDs, Optical Drives	
Molex (4-Pin)	Older Drives, Fans, Accessories	

TYPES OF PSUs

NON-MODULAR	SEMI-MODULAR	FULLY MODULAR
All cables are fixed to the PSU.	Some cables are detachable.	All cables are detachable.
- Lower cost - More cables to manage	- Better cable management - More affordable	- Best cable management - Higher cost

COMMON DC OUTPUTS

RAIL	VOLTAGE	USED FOR
+3.3V	+3.3 Volts	Motherboard, RAM, Chipset
+5V	+5 Volts	USB, HDDs, SSDs, Fans
+12V	+12 Volts	CPU, GPU, Motors, Fans
-12V	-12 Volts	Legacy/Serial ports (rarely used)
+5VSB	+5 Volts Standby	Power for standby features (Wake-on-LAN, etc.)

Note: The +12V rail is the most important for modern PCs as it powers the CPU and GPU.

HOW TO CHOOSE THE RIGHT PSU

- Check your system's power requirements (use a PSU calculator).
- Choose a reputable brand for reliability and safety.
- Pick the right wattage (don't go too low or unnecessarily high).
- Look for 80 PLUS certification for better efficiency.
- Choose the right type (non-modular, semi, or fully modular).

SAFETY PROTECTIONS (BUILT INTO PSUs)

OCP	OVP	UVP	SCP	OPP	OTP
Over Current Protection	Over Voltage Protection	Under Voltage Protection	Short Circuit Protection	Over Power Protection	Over Temp Protection

PRO TIP
A high-quality PSU protects your expensive components and ensures system stability. Never compromise on the power supply!

CAUTION
Using a low-quality or underpowered PSU can cause crashes, data loss, or even damage to your components.

STILL WIDELY USED

Desktops & Workstations	Servers & Data Centers	Gaming PCs	Industrial Systems
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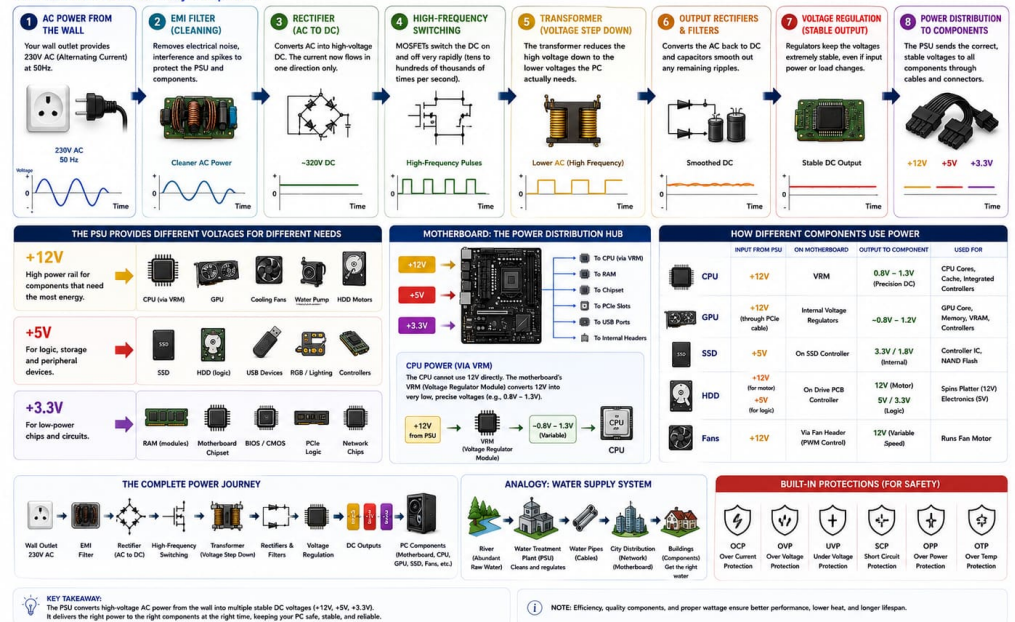
PSUs may not be the most exciting part of a PC, but they are one of the most important!

How a PSU converts and delivers power in a PC

- AC !' rectifier !' transformer !' DC rails.
- Delivers 12 V, 5 V and 3.3 V to the board and drives.
- Undersized PSU = crashes under GPU load.

HOW A PSU CONVERTS AND DELIVERS POWER IN A PC

From the Wall Outlet to Every Component



How a PSU converts and delivers power in a PC

Computer case (chassis)

- Provides airflow, mounting and physical protection.
- Form factors: ATX, Micro-ATX, Mini-ITX.
- Front panel connects power button, USB and audio.

COMPUTER CASE (CHASSIS)

THE FOUNDATION OF YOUR PC

A computer case houses and protects all the internal components of your PC, provides structure, enables airflow for cooling, and determines compatibility, expandability, and aesthetics.

WHAT DOES A CASE DO?

- PROTECTS COMPONENTS**
Shields delicate parts from dust, impacts, and damage.
- SUPPORTS HARDWARE**
Provides mounts and space for motherboard, GPU, drives, PSU, cooling, and more.
- MANAGES AIRFLOW**
Designed with vents and fan mounts to keep components cool.
- ENHANCES BUILD EXPERIENCE**
Good cable management and layout make building and upgrading easier.
- SHOWCASES YOUR STYLE**
From minimal to RGB, your case reflects your personality.

CASE ANATOMY (TYPICAL ATX MID-TOWER)



COMMON CASE TYPES



CASE SIZE GUIDE (BY MOTHERBOARD SUPPORT)

CASE SIZE	MOTHERBOARD SUPPORT	BEST FOR
Full Tower	E-ATX, ATX, mATX, Mini ITX	High-end builds, workstations, custom water cooling
Mid Tower	ATX, mATX, Mini ITX	Gaming, productivity, all-purpose builds
Mini Tower	mATX, Mini ITX	Budget builds, office PCs
Micro ATX	mATX, Mini ITX	Compact builds
ITX / Mini ITX	Mini ITX only	Small form factor builds

MATERIALS & CONSTRUCTION

- STEEL (SPCC)**
Strong, durable, and affordable. Most common.
- ALUMINUM**
Lightweight, premium look and feel. Better for portability.
- TEMPERED GLASS**
Shows off your build. Usually on the side panel.
- PLASTIC**
Used in some panels and front designs. Lighter, less durable.



AIRFLOW & COOLING

OPTIMAL AIRFLOW SETUP (EXAMPLE)



FAN & RADIATOR SUPPORT (EXAMPLE: MID-TOWER)



FRONT I/O PANEL (TYPICAL EXAMPLE)



KEY FEATURES TO CONSIDER

- COMPATIBILITY**
Ensure the case supports your motherboard size, GPU length, CPU cooler height, and PSU size.
- AIRFLOW**
Look for mesh panels, ventilation, and support for multiple fans or radiators.
- EXPANDABILITY**
More drive bays, expansion slots, and fan mounts = more upgrade options.
- CABLE MANAGEMENT**
Good routing space and tie-down points make builds cleaner and airflow better.
- BUILD QUALITY**
Solid construction means durability, less vibration, and better protection.
- AESTHETICS**
Choose a design, color, and features (RGB, glass, minimal, etc.) you love.
- PRO TIP**
A good case with excellent airflow can lower temperatures, reduce noise, and extend the life of your components.

Rear I/O panel — old vs latest

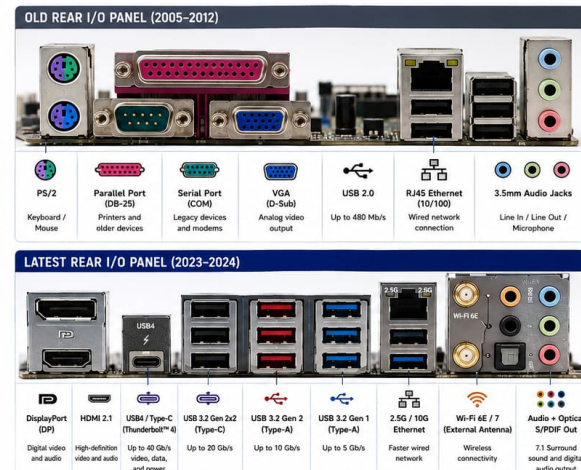
- Old boards: PS/2, VGA, parallel, serial.
- New boards: USB-C, HDMI, DisplayPort, 2.5G Ethernet.
- Rear I/O is soldered — you cannot swap it out.

REAR I/O PANEL OLD VS. LATEST

The Rear I/O (Input/Output) Panel is the section on the back of a computer where you connect external devices such as monitors, keyboards, mouse, USB drives, network cables, audio speakers, and more.

KEY FUNCTIONS

- Connects External Devices**
Allows connection of peripherals such as USB devices, monitors, network, audio, and more.
- Provides Communication**
Enables data transfer between your computer and external devices.
- Expands Capabilities**
Different ports and versions support faster speeds, higher resolutions, and more features.



OLD PANEL HIGHLIGHTS

- Focus on legacy ports (PS/2, Parallel, Serial, VGA)
- USB 2.0 is the fastest available
- Ethernet up to 100 Mb/s
- Analog video and audio
- Limited speed and functionality by today's standards

LATEST PANEL HIGHLIGHTS

- High-speed USB (up to 20 Gb/s and 40 Gb/s)
- Modern video outputs (DP, HDMI 2.1)
- 2.5G / 10G Ethernet for ultra-fast networking
- Wi-Fi 6E/7 for advanced wireless
- High-quality audio with optical output
- More bandwidth, speed, and features for modern devices

QUICK COMPARISON

Feature	Old Panel (2005-2012)	Latest Panel (2023-2024)
USB Version	USB 2.0 (up to 480 Mb/s)	USB 3.2 Gen 2x2 / USB4 (up to 20/40 Gb/s)
Video Outputs	VGA (Analog)	DisplayPort 1.4/2.1, HDMI 2.1
Network	10/100 Ethernet	2.5G / 10G Ethernet, Wi-Fi 6E/7
Audio	3.5mm Analog	High-Definition Audio, Optical S/PDIF
Legacy Ports	PS/2, Parallel, Serial	Removed (Rarely used)
Performance	Slower	Much Faster
Overall	Basic connectivity	High speed, more features, future-ready

EVOLUTION OVER TIME



WHY IT MATTERS

- Speed**
Newer ports offer much faster data transfer.
- Compatibility**
Supports modern devices and higher resolutions.
- Future-Ready**
Latest I/O panels ensure your system stays up to date.
- Better Experience**
Improved audio, network, and connectivity for gaming, streaming, and productivity.

REAR I/O PANELS CONTINUE TO EVOLVE, BRINGING MORE SPEED, BETTER CONNECTIVITY, AND ENHANCED FEATURES TO MEET MODERN DEMANDS.

Rear I/O panel — old vs latest

New ports and connections — modern connectivity

- USB-C is reversible and delivers data + video + power.
- Thunderbolt 4/5 is the fastest general port on a PC.
- Learn what each port can and can't carry.

NEW PORTS & CONNECTIONS

MODERN PORTS FOR TODAY'S DEVICES

Technology is always evolving. New ports and connections deliver faster speeds, more power, higher video resolutions, and greater versatility than ever before.

KEY BENEFITS

- Faster Data Transfer**
Move large files in seconds.
- Higher Video & Audio Quality**
Support for 4K, 8K and beyond.
- More Power Delivery**
Charge laptops and devices with a single cable.
- Universal Connectivity**
One port for many devices and uses.
- Future-Ready**
Built for modern devices and tomorrow's demands.

LATEST PORTS & CONNECTIONS OVERVIEW

USB4® (40Gb/s)	THUNDERBOLT™ 4 (40Gb/s)	USB 3.2 GEN 2x2 (20Gb/s)	USB-C® (USB 3.x / USB4)	HDMI® 2.1	DISPLAYPORT™ 2.1	2.5G / 5G / 10G ETHERNET	WI-FI 6E / WI-FI 7 (BLUETOOTH® 5.3+)
• Up to 40 Gb/s • Data, video, audio • Power Delivery (up to 100W) • Compatible with Thunderbolt 3/4	• Up to 40 Gb/s • Dual 4K or single 8K video support • Power Delivery (up to 100W) • Daisy-chain up to 6 devices	• Up to 20 Gb/s • Backward compatible with USB 3.2/2.0 • Ideal for fast external storage	• Reversible connector • Data, video, audio • Power Delivery • Many protocols in one small port	• Up to 48 Gbps • 4K @ 120Hz, 8K @ 60Hz • Dynamic HDR, eARC • VRR, ALLM, QFT for gaming	• Up to 80 Gbps • Up to 16K @ 60Hz • DSC, HDR, VRR, MST • High refresh rate support	• 2.5 / 5 / 10 Gbps speeds • Backward compatible with 1G/100M • Better for streaming, gaming & transfers	• 6 GHz band (6E) or Multi-Link Operation (7) • Lower latency • Higher throughput • Bluetooth 5.3+ for better range & audio

WHAT CAN THEY CONNECT?

USB4 / Thunderbolt 4	USB 3.2 Gen 2x2	USB-C (PD)	HDMI 2.1	DisplayPort 2.1	2.5G / 5G / 10G Ethernet	Wi-Fi 6E / Wi-Fi 7
Docks, monitors, external GPUs, storage, eGPUs, network adapters, and more.	Fast external SSDs/HDDs, cameras, smartphones, 10Gb adapters.	Laptops, tablets, phones, power banks, and accessories with power delivery.	TVs, monitors, projectors, soundbars, AV receivers, consoles, streaming devices.	High-resolution monitors, VR headsets, professional displays, daisy-chained setups.	Routers, switches, NAS, network attached storage, internet connections.	Wireless routers, smart home devices, AR/VR, Bluetooth audio & more.

COMMON CONNECTOR SHAPES

USB-A (3.x)	USB-C	Micro-USB	Mini-DisplayPort	HDMI	DisplayPort
RJ45 Ethernet	3.5mm Audio Jack	Optical Audio (TOSLINK)	USB-B (3.x)	USB-C (PD)	Thunderbolt 4

PORT SPEED COMPARISON

Port Type	Speed (Gbps)
USB 2.0	480 Mbps
USB 3.2 Gen 1	5 Gbps
USB 3.2 Gen 2	10 Gbps
USB 3.2 Gen 2x2	20 Gbps
USB4 / TB4	40 Gbps
Thunderbolt 5 (Upcoming)	Up to 120 Gbps

QUICK TIPS

- Always use certified cables for best performance.
- USB-C ports can support different features. Check your device specifications.
- For high refresh rates and resolutions, use HDMI 2.1 or DisplayPort 1.4/2.1.
- Use wired Ethernet for the most stable and fastest internet connection.
- Keep your drivers and firmware up to date for maximum compatibility.

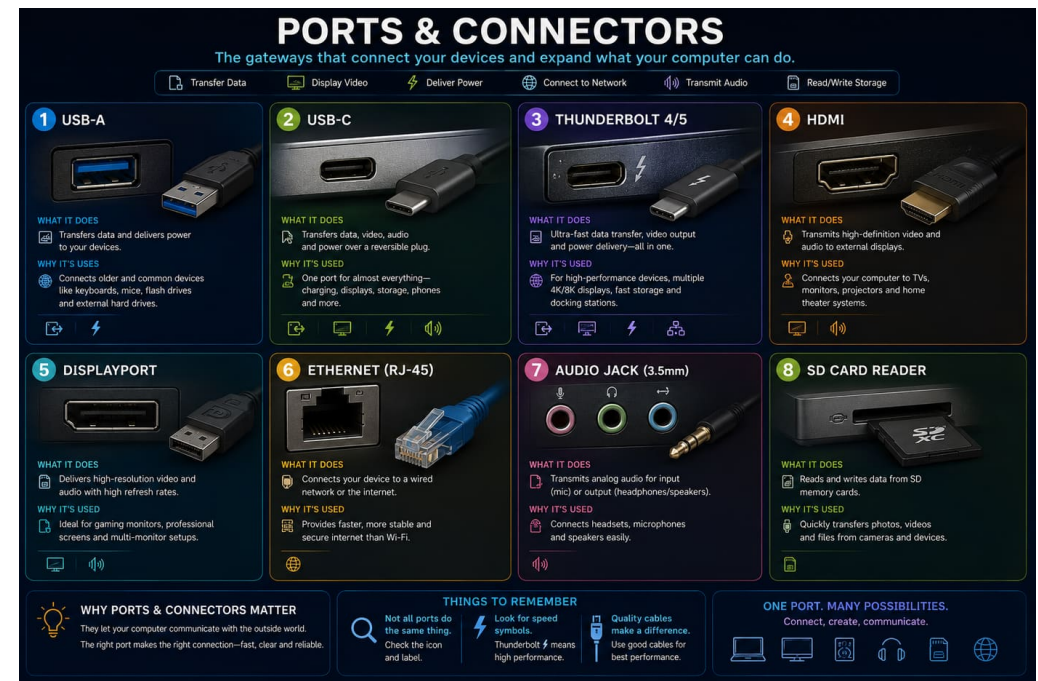
ONE PORT CAN DO IT ALL!

Modern ports like USB-C, USB4, and Thunderbolt 4 combine data, video, audio, and power in a single connection—making your setup cleaner, faster, and more powerful.

SSP+ = SuperSpeed USB PD = Power Delivery VRR = Variable Refresh Rate
 eARC = Enhanced Audio Return Channel MST = Multi-Stream Transport

Ports and connectors — USB-A/C, Thunderbolt, HDMI, DisplayPort, Ethernet, audio, SD

- Each port has a specific role: data, video, network, audio.
- Not all USB-C ports carry video or Thunderbolt.
- Cable quality matters — cheap cables fail silently.



Expansion hardware — add-in cards

- PCIe cards add Wi-Fi, capture, sound, extra USB, RAID.
- Slot size: x1, x4, x8, x16 — must fit the card.
- Great way to modernise an older machine.

EXPANSION HARDWARE

Add more features, performance, and connectivity to your computer.

WHAT IS EXPANSION HARDWARE?

Expansion hardware are additional cards or modules installed in a computer to add or enhance capabilities beyond the built-in features of the motherboard.

KEY

- PCle Slot
- M.2 Slot
- USB Header / Internal

1. WI-FI CARD

Adds wireless internet connectivity to your PC.

Interface: PCIe x1

- Wi-Fi 6 / Wi-Fi 6E / Wi-Fi 7
- High-speed wireless
- Better range and stability
- Supports 2.4GHz, 5GHz and 6GHz bands

Common Uses

- Wireless Internet
- Online Gaming
- Streaming
- Large File Downloads

2. BLUETOOTH MODULE

Adds Bluetooth connectivity to your PC.

Interface: M.2 (Key E) or USB

- Bluetooth 5.0 / 5.2 / 5.3+
- Connects mouse, keyboard, headphones, speakers, etc.
- Low power consumption
- Compact and easy to install

Common Uses

- Wireless Peripherals
- Headphones / Audio
- File Transfer
- IoT / Smart Devices

3. CAPTURE CARD

Captures video and audio from another device.

Interface: PCIe x4 (or x1)

- Record or stream gameplay
- Supports HDMI / SDI / Component / Composite
- 4K / 1080p high-quality
- Low latency passthrough

Common Uses

- Game Streaming
- Video Recording
- Live Broadcasting
- Content Creation

4. SOUND CARD

Enhances or adds high-quality audio to your system.

Interface: PCIe x1

- Better audio quality
- 7.1 Surround sound
- Lower latency
- Dedicated audio processing

Common Uses

- Gaming
- Music Production
- Home Theater
- Video Recording

5. RAID CONTROLLER

Manages multiple drives for better performance or redundancy.

Interface: PCIe x4 / x8

- RAID 0, 1, 5, 10, etc.
- Hardware RAID (offloads CPU)
- More reliable data storage
- Supports SAS / SATA drives

Common Uses

- Servers
- Data Protection
- Workstations
- High-Speed Storage

6. THUNDERBOLT EXPANSION CARD

Adds Thunderbolt ports for ultra-fast data, video, and power.

Interface: PCIe x4

- Thunderbolt 3 / 4 support
- Up to 40 Gbps bandwidth
- Supports 4K/8K displays, docks, NVMe drives
- Daisy-chain multiple devices

Common Uses

- High-Speed Storage
- 4K / 8K Monitors
- Thunderbolt Docks
- Professional Workflows

7. USB EXPANSION CARD

Adds more USB ports to your PC.

Interface: PCIe x1

- USB 3.0 / 3.1 / 3.2 / 3.2 Gen 2x2
- Add multiple USB ports
- Supports front panel USB
- Internal header support

Common Uses

- More USB Ports
- Peripherals
- Front Panel Connections
- Data Transfer

HOW EXPANSION CARDS CONNECT

EXPANSION SLOTS & CONNECTORS

- PCle** (Peripheral Component Interconnect Express): Used for high-performance cards like graphics cards, RAID controllers, sound cards, and more.
- PCle M.2 Slot (Key E / Key M)**: Used for smaller modules like Wi-Fi, Bluetooth, and NVMe SSDs.
- PCle USB Header (Internal)**: Connects USB ports to the front panel or adds internal USB support.

BENEFITS OF EXPANSION HARDWARE

- ✓ Extend system capabilities
- ✓ Improve performance
- ✓ Add modern connectivity
- ✓ Customise your PC for your needs
- ✓ Future-proof your system

TIPS

- Check compatibility (slot type, size, bandwidth).
- Install the latest drivers for best performance.
- Use quality cards from reputable brands.
- Follow ESD safety practices during installation.

Input devices — the essentials

- Keyboard, mouse, touchpad, touchscreen.
- Webcam and microphone for calls and content.
- Fingerprint reader for secure sign-in.



Input devices — the extended catalogue (24 devices)

- Stylus, trackball, joystick, controller, light gun.
- Scanner, barcode/QR reader, smart card reader.
- MIDI, eye tracker, voice, iris, data glove, foot pedal.



Output devices — monitors, speakers, headphones, projectors, VR, printers and more

- Sight: monitors, projectors, VR headsets, smart glasses, LED displays.
- Hearing: speakers and headphones.
- Touch & other senses: printers, plotters, 3D printers, haptic gloves, braille displays.



Networking hardware (2026)

- Router: connects the LAN to the Internet.
- Switch: connects wired devices inside the LAN.
- Access point: provides Wi-Fi coverage.

NETWORKING HARDWARE 2026

The Latest & Best Devices – What They Do and Why They Matter

The backbone of a fast, secure and always-connected world.

1	2	3	4	5	6	7
ETHERNET CONTROLLER	NETWORK INTERFACE CARD (NIC)	WI-FI 7 ADAPTER	NETWORK SWITCH	ROUTER	MODEM	ACCESS POINT
						
Built into the motherboard or server.	Add-on card that connects a device to a network.	Connects your device to wireless networks.	Connects multiple devices within a network.	Connects your network to the internet and other networks.	Connects your internet service to your network.	Creates Wi-Fi coverage for wireless devices.
WHAT IT DOES Handles wired network communication between your device and the network.	WHAT IT DOES Adds high-speed wired connectivity to desktops, servers or workstations.	WHAT IT DOES Delivers ultra-fast Wi-Fi, lower latency and more stable connections.	WHAT IT DOES Directs data between devices on a local network (LAN) quickly and efficiently.	WHAT IT DOES Routes data between different networks and protects your network with built-in security.	WHAT IT DOES Converts the internet signal from your ISP (cable, fiber, 5G, DSL) so your network can use it.	WHAT IT DOES Broadcasts Wi-Fi signal and connects many wireless devices seamlessly.
WHY IT'S USED Provides fast, reliable and efficient data transfer for everything connected to a network.	WHY IT'S USED Boosts performance, adds advanced features and more reliability than built-in ports.	WHY IT'S USED Perfect for gaming, streaming, AR/VR, remote work and smart homes.	WHY IT'S USED Keeps your network organized, fast and running smoothly as your devices grow.	WHY IT'S USED Shares internet securely, manages traffic and connects offices or branches together.	WHY IT'S USED Provides the link between your home or office and your internet provider.	WHY IT'S USED Expands Wi-Fi coverage in homes, offices and large spaces with strong, reliable signal.
EXAMPLE (2026) Intel Ethernet 800 Series	EXAMPLE (2026) NVIDIA ConnectX-7	EXAMPLE (2026) Intel BE200 Wi-Fi 7 Adapter	EXAMPLE (2026) Cisco Catalyst 9600X Series	EXAMPLE (2026) Cisco Catalyst 8300 Series	EXAMPLE (2026) Arris SURFboard S34 DOCSIS 4.0	EXAMPLE (2026) HPE Aruba AP-775 Wi-Fi 7 (802.11be)

WHY THIS HARDWARE MATTERS TO YOU

SPEED

Get faster downloads, smarter streaming and smoother gaming.

RELIABILITY

Stay connected with stable, consistent performance.

SECURITY

Protect your data with built-in security and threat defense.

SCALABILITY

Easily support more devices, users and future growth.

PERFORMANCE

Built to handle today's demands and tomorrow's innovations.

EFFICIENCY

Lower latency, less power usage, better experience.

BUILT FOR THE FUTURE

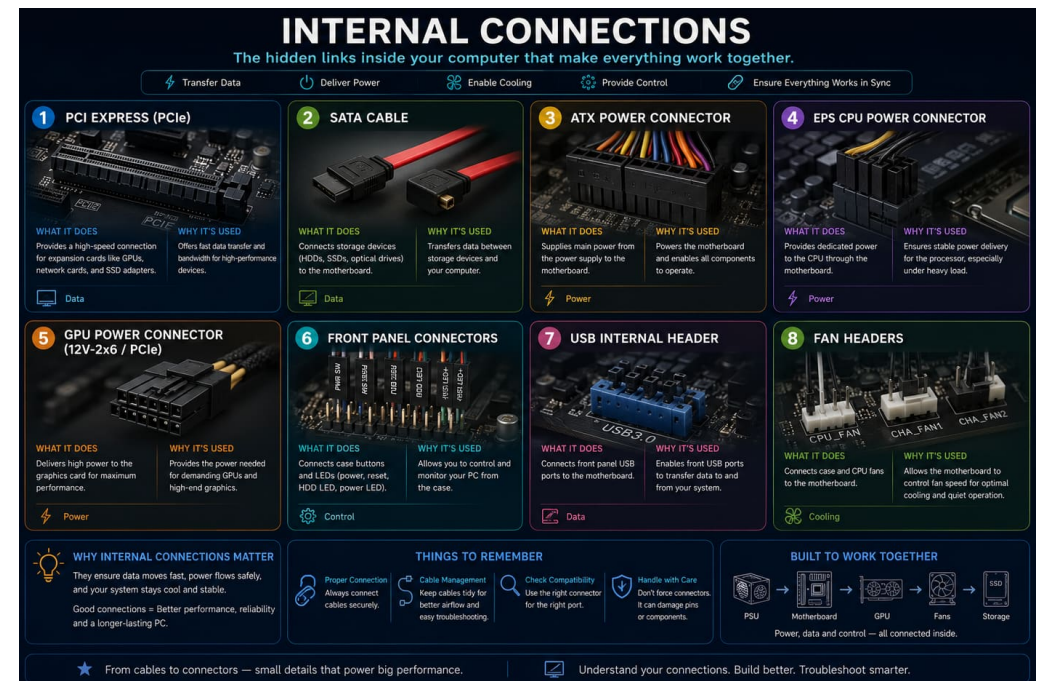
Wi-Fi 7, 10G/25G/100G speeds, AI-driven networks, and advanced security make these devices future-ready.

Together, these devices create a powerful network that keeps you connected, productive and secure.

For homes, businesses, schools and data centers.

Networking and connectivity — how devices reach each other

- Wired vs wireless: speed vs convenience.
- IP addresses identify devices on the network.
- DNS turns names into IP addresses.

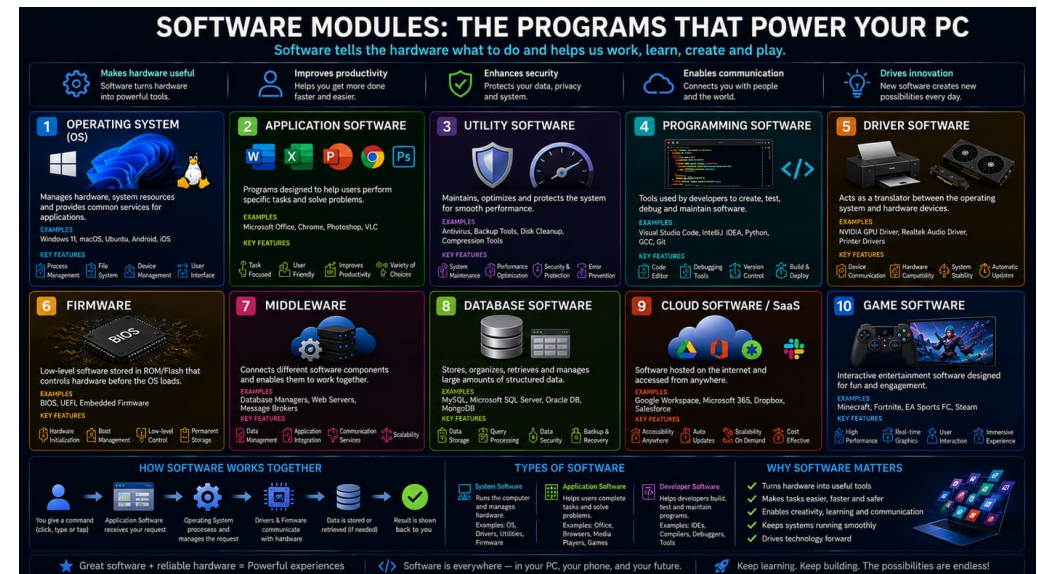


PART 2 — SOFTWARE: the programs that bring the hardware to life

Every slide in Part 2 is code — from firmware and the operating system, up through applications, and finally the modern layer of cloud, virtualisation, networking, security and AI software.

Software modules — the programs that power your PC

- OS, apps, utilities, drivers, firmware, middleware.
- Databases, cloud, games — every type has a role.
- Software tells hardware what to do.



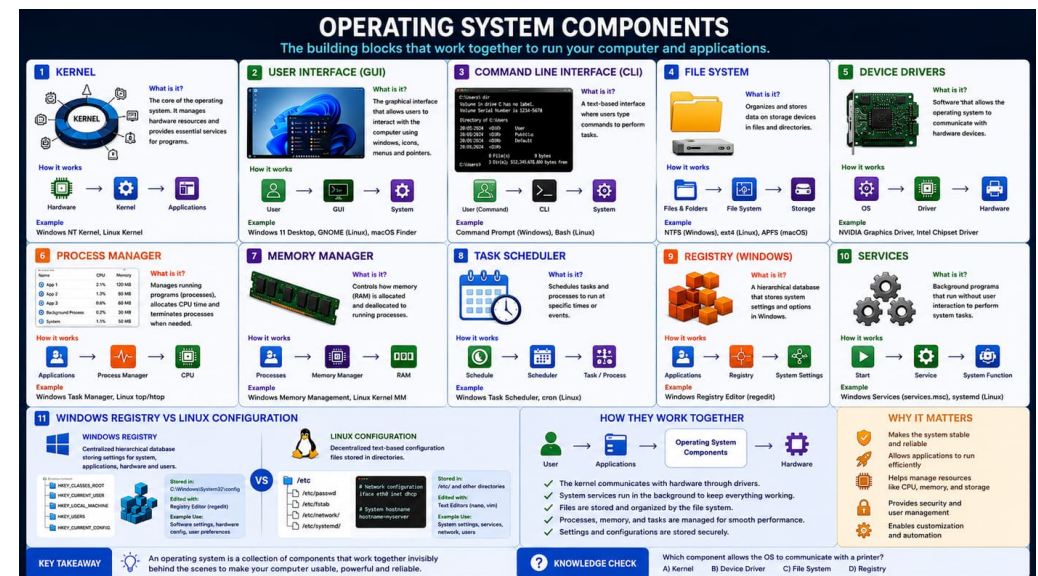
Operating systems — the core system software

- Windows, macOS, Linux, ChromeOS, Android, iOS.
- Manages hardware, users, files, security.
- Every app depends on the OS to run.



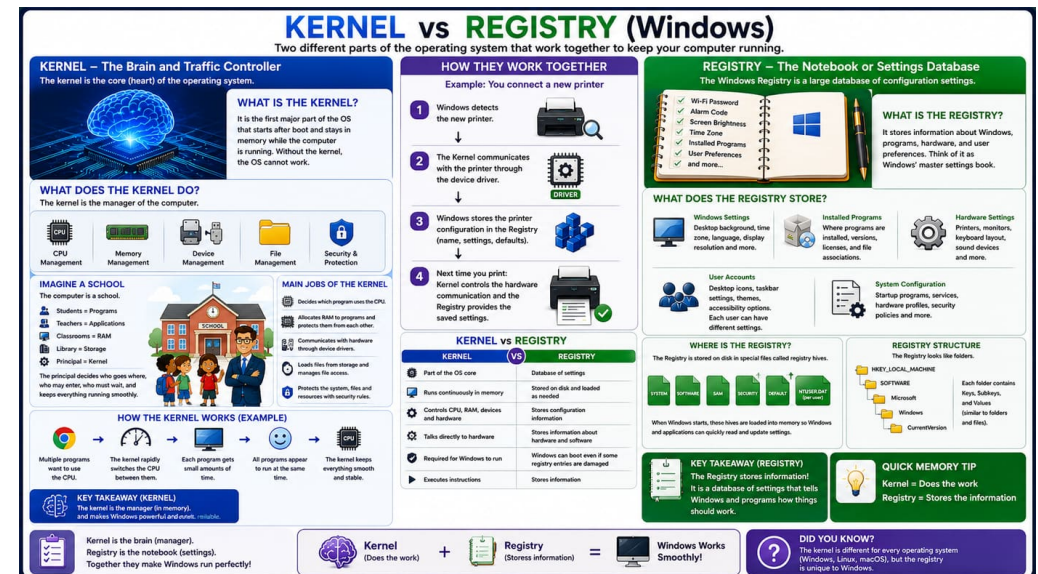
Operating system components — kernel, drivers, file system, services, UI

- Kernel talks to hardware; UI talks to the user.
- Drivers translate between the two.
- File system organises everything on disk.



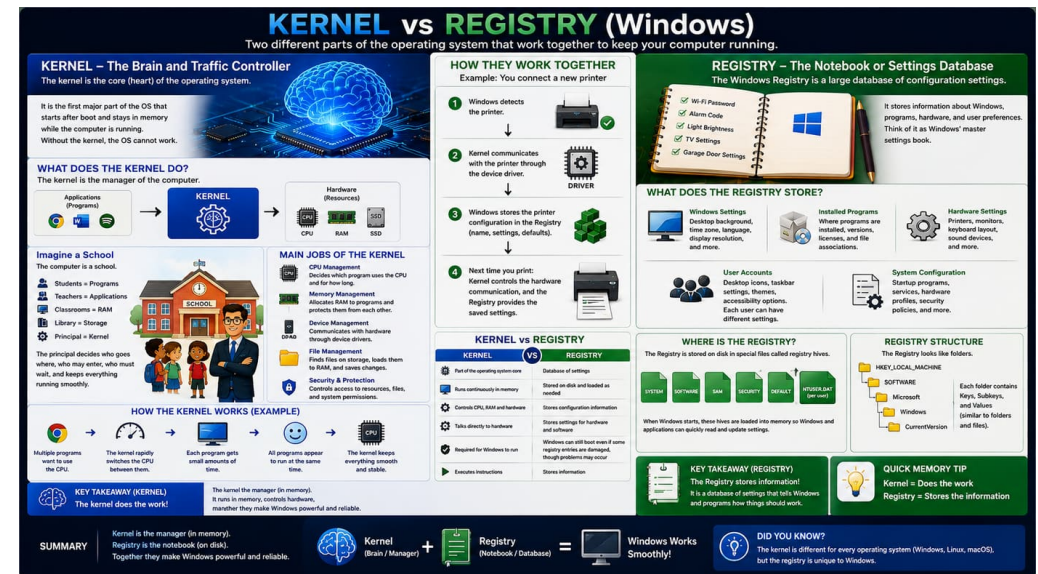
Kernel vs Registry on Windows (part 1)

- Kernel = the core code that runs the OS.
- Registry = a database of settings for OS and apps.
- Different jobs, often confused.



Kernel vs Registry on Windows (part 2)

- Kernel bugs can crash the whole machine (BSOD).
- Registry edits can break login or app behaviour.
- Back up before touching either.



When would you actually want to use the Registry?

- Applying a fix that has no GUI setting.
- Removing stubborn leftover app entries.
- Enterprise policy changes via Group Policy.



When would you actually want to use the Registry?

Firmware and low-level software

- Firmware lives inside chips (BIOS/UEFI, SSD, NIC).
- Updated with vendor tools — carefully.
- Bad firmware update can brick a device.

FIRMWARE & LOW-LEVEL SOFTWARE 2026

The invisible software that powers and controls hardware.

Firmware is specialized software embedded in hardware that provides core control, initializes devices, and enables communication between hardware and operating systems.

WHY IT MATTERS

- Enables & Controls Hardware
- Security & Trust Foundation
- Performance & Stability
- Updates & Compatibility
- Critical for Business Continuity

1 BIOS Basic Input/Output System Legacy firmware that initializes hardware and starts the boot process. Key Functions: - Power-On Self Test (POST) - Hardware initialization - Boot device selection - Basic system configuration	2 UEFI Unified Extensible Firmware Interface Modern firmware interface that replaces BIOS with advanced features and security. Key Functions: - Faster boot and shutdown - Supports large drives (>2TB) - Secure Boot and TPM integration - Graphical interface & mouse support	3 Device Firmware (General) Firmware embedded in individual devices to control their specific functions. Key Functions: - Device initialization & control - Manage device features - Ensure hardware compatibility - Provide bug fixes & improvements	4 SSD Firmware Solid State Drive Firmware Controls SSD operations, flash memory management and data reliability. Key Functions: - Wear leveling & garbage collection - Error correction & data integrity - Performance optimization - Self-monitoring (SMART)	5 GPU Firmware (VBIOS) Video BIOS Firmware on the graphics card that controls hardware operation and settings. Key Functions: - Initialize GPU hardware - Manage clocks, voltages & fans - Display output configuration - Ensure stability & compatibility
6 Motherboard Firmware (Board Firmware) Firmware that manages the motherboard's components and built-in devices. Key Functions: - Chipset & component control - Power management - Hardware monitoring - Feature enablement & updates	7 TPM Firmware Trusted Platform Module Firmware Firmware that runs on the TPM chip to provide hardware-based security. Key Functions: - Secure key generation & storage - Platform integrity measurement - Encryption & authentication - Secure boot support	8 Embedded Controllers (EC Firmware) Firmware in embedded controllers that manage low-level system functions. Key Functions: - Power button & sleep control - Keyboard, fan & thermal control - Battery & charging management - Sensor monitoring	9 Secure Boot (UEFI Feature) Security feature that ensures only trusted firmware and OS components run. Key Functions: - Verifies digital signatures - Blocks malicious bootloaders - Protects boot process integrity - Works with TPM for trust	10 Firmware Updates Updates to firmware that add features, fix bugs and close security issues. Key Functions: - Bug fixes & stability improvements - Security vulnerability patches - Performance enhancements - Hardware compatibility updates

ACRONYM EXPLANATIONS

BIOS - Basic Input/Output System	TPM - Trusted Platform Module
UEFI - Unified Extensible Firmware Interface	POST - Power-On Self Test
SSD - Solid State Drive	SMART - Self-Monitoring, Analysis and Reporting Technology
GPU - Graphics Processing Unit	
VBIOS - Video BIOS (Firmware on GPU)	

COMMONLY USED BY ORGANIZATIONS LIKE INVESTEC FOR

System Security	Hardware Performance	Data Integrity	System Reliability	Regulatory Compliance
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BEST PRACTICES

- Keep firmware up to date from official sources.
- Enable Secure Boot and TPM for stronger security.
- Test updates in a controlled environment first.
- Monitor release notes for fixes and known issues.
- Ensure power stability during firmware updates.

IMPORTANT: Incorrect firmware updates can cause system instability or hardware failure. Always follow vendor instructions and backup configurations.

PRO TIP: Regular firmware maintenance improves security, stability, performance and hardware lifespan.

2026 Stay updated. Stay secure. Stay productive.

Application software — the big picture

- Programs users interact with directly.
- Run on top of the OS.
- Often licensed per user or per device.



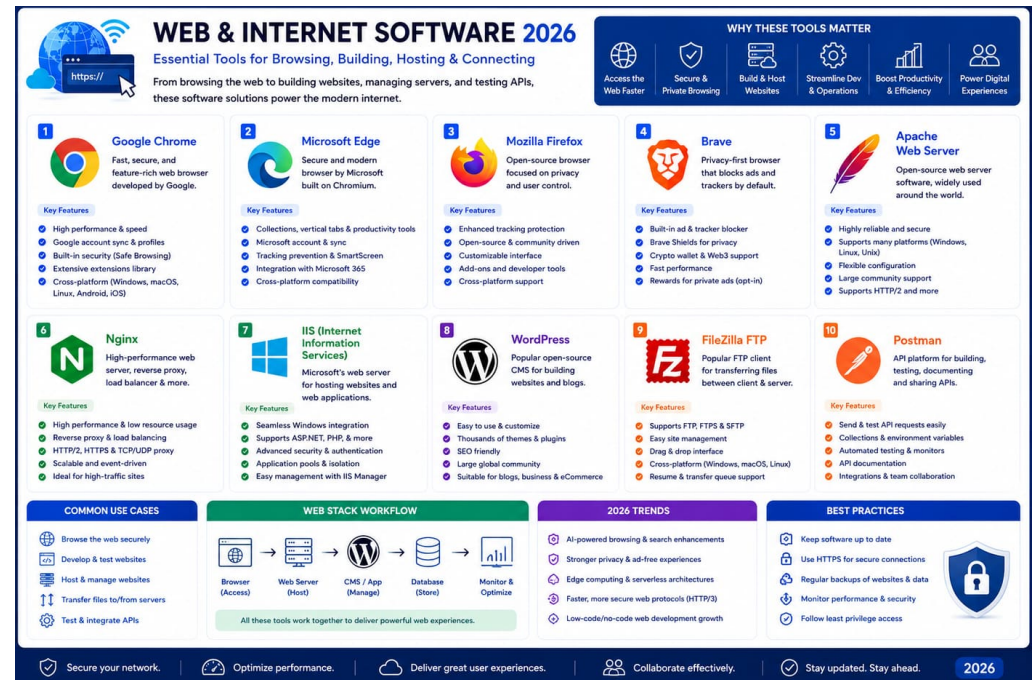
Enterprise and business software (ERP, CRM, HR)

- ERP: finance, stock, procurement (SAP, Oracle).
- CRM: customer records (Salesforce, Dynamics).
- HR: people and payroll (Workday, Sage).



Web and internet software

- Browsers, email clients, messaging apps.
- Web servers, CMS platforms.
- Most 'apps' today are really web apps.



Multimedia and creative software

- Photo, video, audio editors (Photoshop, Premiere).
- 3D and design tools (Blender, AutoCAD).
- GPU-hungry — plan hardware accordingly.

MULTIMEDIA & CREATIVE SOFTWARE 2026
Powering Creativity. Inspiring Ideas. Delivering Impact.

From photo editing and vector design to video production, animation, audio recording and live streaming—these tools help creators bring their ideas to life.

WHY THESE TOOLS MATTER

- Unleash Creativity
- Boost Productivity
- Enhance Collaboration
- Save Time & Increase Efficiency
- Create Professional Quality Content
- Grow Your Brand & Audience

- 1 Adobe Photoshop**
Industry-standard tool for photo editing and raster graphics.
KEY FEATURES
 - Advanced photo editing & retouching
 - Layers, masks & smart objects
 - AI-powered tools (Generative Fill, etc.)
 - Wide range of filters & effects
 - Supports RAW & 3D editing
- 2 Adobe Illustrator**
Professional vector graphics software for illustrations, logos and typography.
KEY FEATURES
 - Vector illustration & design
 - Scalable artwork without quality loss
 - Advanced typography tools
 - Artboards & responsive assets
 - Integration with Adobe Creative Cloud
- 3 Adobe Premiere Pro**
Professional video editing software for film, TV and online content.
KEY FEATURES
 - Multi-track video editing
 - High-performance playback
 - Color correction & audio tools
 - 3D animation & camera tools
 - AI-powered features (Auto Reframe)
 - Seamless Adobe ecosystem integration
- 4 Adobe After Effects**
Motion graphics & visual effects for stunning animations and compositions.
KEY FEATURES
 - Motion graphics & visual effects
 - Keying, tracking & compositing
 - Hundreds of effects & presets
 - Integration with Premiere Pro & more
- 5 Audacity**
Free, open-source audio editor and recorder.
KEY FEATURES
 - Record & edit audio easily
 - Multi-track editing
 - Noise reduction & audio restoration
 - Supports many audio formats
 - Plugins & effect support
- 6 OBS Studio**
Free and open-source software for live streaming and screen recording.
KEY FEATURES
 - Live streaming to multiple platforms
 - Scene & source management
 - Real-time audio mixing
 - High performance & lightweight
 - Plugins & extensive customization
- 7 Blender**
Powerful open-source 3D creation suite for modeling, animation, rendering and more.
KEY FEATURES
 - 3D modeling, sculpting & animation
 - Rendering with Cycles & Eevee
 - Simulation (physics, fluids, smoke)
 - Video editing & VFX
 - Completely free & open-source
- 8 DaVinci Resolve**
Professional video editing, color grading, visual effects and audio post-production.
KEY FEATURES
 - Advanced color correction & grading
 - Edit, VFX, motion graphics & audio in one tool
 - High-end professional quality
 - Collaboration & cloud workflows
 - Free version available (powerful)
- 9 Canva**
User-friendly design platform for graphics, presentations, social media and more.
KEY FEATURES
 - Drag-and-drop design tools
 - Thousands of templates & assets
 - Brand kits & team collaboration
 - Video & animation maker
 - Web, print & social content support
- 10 Figma**
Collaborative UI/UX design tool for teams and digital products.
KEY FEATURES
 - Real-time collaboration
 - UI/UX design & prototyping
 - Design systems & components
 - Developer handoff & inspect mode
 - Cloud-based & platform-independent

COMMON USE CASES

- Photo Editing & Retouching
- Graphic Design & Illustration
- Video Editing & Production
- Motion Graphics & Visual Effects
- Audio Recording & Editing
- Live Streaming & Webcasts

KEY BENEFITS

- Enhances Creativity
- Saves Time & Increases Efficiency
- Professional Quality Output
- Better Collaboration & Workflow
- Cost-Effective Solutions

2026 TRENDS

- AI-powered creative assistance
- Cloud collaboration & real-time workflows
- Higher quality, 8K & immersive content
- Cross-platform & integrated ecosystems
- Creator economy & digital storytelling growth

CREATIVE WITH PURPOSE
The right tools. The right ideas. The right impact.

KEEP LEARNING. KEEP CREATING. KEEP INSPIRING.
Master the tools. Share your vision. Shape the future.

2026
Create today. Inspire tomorrow.

Database software

- Stores structured data (SQL Server, MySQL, Postgres).
- Queried with SQL.
- Backups are critical — data loss = job loss.

DATABASE SOFTWARE
Store. Manage. Protect. Analyze.
Database software helps organizations store, manage, retrieve and analyze data efficiently and securely.

WHAT DATABASE SOFTWARE DOES

- STORE**: Organize data efficiently
- MANAGE**: Create, update and maintain
- RETRIEVE**: Search and access data
- ANALYZE**: Generate insights and reports
- PROTECT**: Ensure security and reliability

POPULAR DATABASE SOFTWARE

- Microsoft SQL Server**: Enterprise-grade relational database from Microsoft.
 - High performance & scalability
 - Advanced security
 - Business intelligence
 - Integration with Microsoft ecosystem
 Best for: Enterprise applications, data warehousing, BI solutions.
- MySQL**: Popular open-source relational database.
 - Fast, reliable and easy to use
 - Cost-effective
 - Large community support
 - Supports SQL standards
 Best for: Web applications, SaaS, startups.
- PostgreSQL**: Advanced open-source relational database.
 - ACID compliant
 - Extensible and powerful
 - Supports complex queries
 - Strong security
 Best for: Complex applications, analytics, GIS, developers.
- ORACLE DATABASE**: Enterprise-grade database for mission-critical workloads.
 - High availability
 - Advanced security
 - Powerful performance
 - Comprehensive tools
 Best for: Large enterprises, mission-critical systems.
- MongoDB**: Leading NoSQL document database.
 - Flexible document model
 - High scalability
 - Schema-less design
 - Fast development
 Best for: Big data, agile apps, real-time applications.
- SQLite**: Lightweight, serverless relational database.
 - No server required
 - Zero configuration
 - Small, fast, reliable
 - ACID compliant
 Best for: Mobile apps, embedded systems, IoT.
- Redis**: In-memory key-value data structure store.
 - Extremely fast
 - Caching and sessions
 - Publish/subscribe messaging
 - High availability
 Best for: Caching, real-time analytics, session management.

DBMS (DATABASE MANAGEMENT SYSTEMS)
A DBMS is software that acts as an interface between users/applications and the database. It helps you define, create, manipulate, control and secure data.

- Define**: Create database structures
- Manipulate**: Insert, update, delete data
- Control**: Manage access and permissions
- Monitor**: Track performance and usage
- Backup & Recover**: Protect and restore data
- Multi-user Support**: Allow multiple users to work safely

SQL vs NoSQL

SQL (Relational)	NoSQL (Non-Relational)
Data Model : Tables (rows & columns)	Documents, key-value, graphs, wide-columns
Schema : Fixed schema	Flexible / Schema-less
Query Language : SQL	Varies (JSON, Queries, etc.)
Relationships : Joins, foreign keys	Embedded or referenced
Scalability : Vertical scaling	Horizontal scaling
Best For : Structured data, complex transactions	Big data, high scale, real-time apps

DBMS FEATURES

- Data Storage
- Data Security
- Data Integrity
- Backup & Recovery
- User Management
- Query Processing
- Performance Optimization

SQL vs NoSQL - WHEN TO USE WHAT?

Use SQL When:

- ✓ You need ACID compliance
- ✓ Your data is structured
- ✓ You need complex transactions
- ✓ You have strict consistency needs

Use NoSQL When:

- ✓ You need high scalability
- ✓ Your data is unstructured or semi-structured
- ✓ You need fast development
- ✓ You handle large volumes of data

COMMON USE CASES

- Web Applications
- E-commerce Platforms
- Business Intelligence
- Mobile Apps
- IoT & Devices
- Banking & Finance
- Healthcare
- Gaming

DID YOU KNOW?
The global database software market is growing rapidly, driven by cloud computing, AI analytics, and real-time data needs.

TRENDS TO WATCH

- Cloud Databases
- AI-Powered Databases
- Real-time Analytics
- Multi-model Databases
- Advanced Security

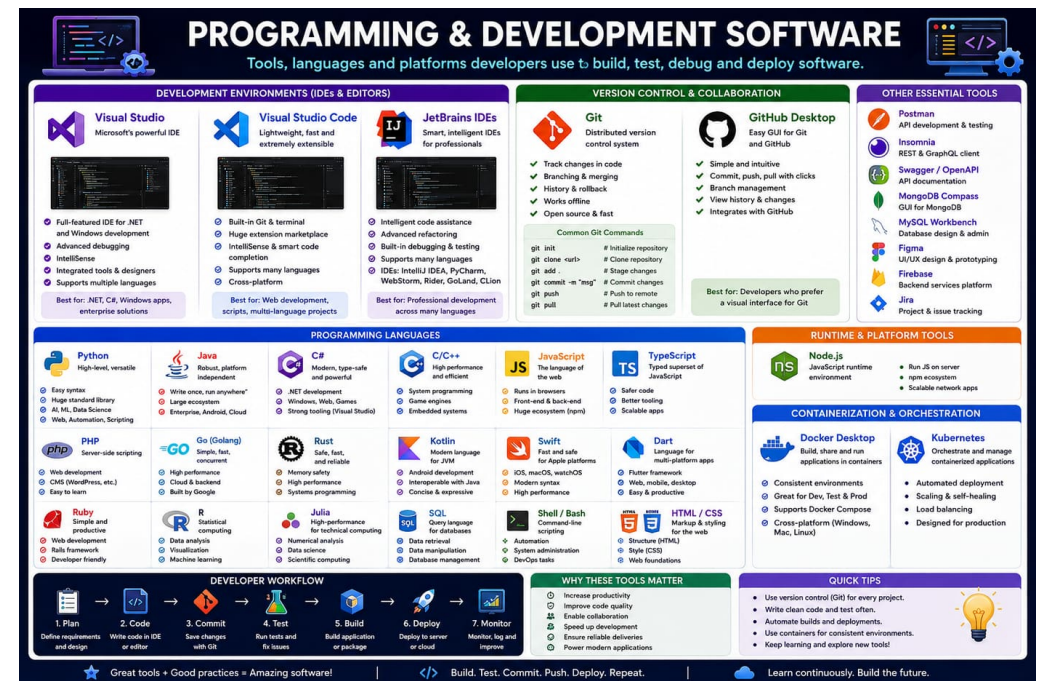
BEST PRACTICES

- Choose the right database for your needs
- Normalize data (SQL) / Model properly (NoSQL)
- Index your data for faster queries
- Regularly backup your databases
- Monitor performance and optimize queries
- Keep database software up to date

Better Data. Better Decisions. Stronger Security. Smarter Future.
Choose the right database. Build the right future.

Programming and development software

- Editors and IDEs (VS Code, IntelliJ, Xcode).
- Compilers, debuggers, version control (Git).
- Where every other software category starts.



Utility software — the small tools that keep systems healthy

- Antivirus, backup, disk clean-up, compression.
- Small tools, big impact on reliability.
- Support technicians rely on utilities daily.

UTILITY SOFTWARE 2026
Essential tools that keep modern organizations secure, efficient and productive.
Used across IT, security, finance and operations by leading companies like Investec to maintain systems, protect data, automate tasks and improve performance.

WHY UTILITY SOFTWARE MATTERS

- Improves Performance
- Protects Data & Systems
- Saves Time & Automates Tasks
- Simplifies IT Management
- Ensures Compliance & Reliability
- Reduces Costs & Optimizes Resources

1 Microsoft PowerToys Utilities to customize Windows for power users. - Advanced keyboard manager - Window layouts & shortcuts - File Explorer add-ons - Power rename, image resizer	2 Microsoft Sysinternals Suite Advanced system utilities for IT pros. - Process Explorer - Autoruns - Symptom - TCPView & more	3 Notepad++ Advanced text editor for developers and IT professionals. - Syntax highlighting - Code folding - Multi-language support - Lightweight & fast	4 Everything Fast file search tool for Windows. - Instant search results - Lightweight - Search by name or content - Filters & advanced search	5 TeamViewer Remote access and remote support. - Remote control devices - File transfer - Unattended access - Cross-platform	6 AnyDesk Fast, secure remote desktop software. - Low latency - Easy to use - File transfer - Cross-platform
7 Veeam Backup & Replication Data backup and disaster recovery solution. - Virtual, physical & cloud backup - Instant recovery - Data deduplication - High availability	8 SolarWinds Toolset Network & system management tools. - Network monitoring - Performance analysis - Configuration management - IT infrastructure visibility	9 ManageEngine ADManager Plus Active Directory management and reporting. - User management - Password reset - AD reporting - Change tracking	10 Malwarebytes Advanced malware detection and removal. - Real-time protection - Ransomware protection - Adware & PUP removal - Threat scanning	11 ESET PROTECT (Endpoint Security) Endpoint protection for businesses. - Antivirus & antispware - Device control - Web & email protection - Centralized management	12 Acronis Cyber Protect Cloud Backup and cybersecurity in one integrated solution. - Image backup - Anti-malware protection - Cloud storage - Disaster recovery
13 WinSCP Secure file transfer (SFTP, SCP, FTP). - Secure file transfers - File management - Script automation - Integration with PuTTY	14 PuTTY SSH and Telnet client for remote connections. - Secure SSH connections - Lightweight - Cross-platform - Scripting support	15 FileZilla FTP client for fast and reliable file transfers. - Easy site management - Resume transfers - Cross-platform - Drag-and-drop interface	16 BleachBit Cleans unnecessary files and frees up space. - System cleaner - Privacy protection - Browser extension clearing - Open source	17 Revo Uninstaller Pro Advanced uninstall and system cleanup. - Complete program removal - Leftover file scanning - Browser extension removal - Forced uninstall	18 Advanced IP Scanner Fast network scanner and IP discovery. - Scan IP ranges - Detect devices - Remote access - Export results
19 SpaceSniffer Disk space analyzer for Windows. - Visualize disk usage - Find large files - Easy to use - Portable	20 HashTab Calculate and verify file hashes. - Supports MD5, SHA-1, SHA-256 - File integrity verification - Right-click integration - Lightweight	21 Print Management (Windows) Manage printers and print servers. - Deploy printers - Monitor print queues - Set permissions - Track usage	22 Windows Task Scheduler Automate tasks and run programs. - Schedule tasks - Triggers & actions - System maintenance - Improves productivity	23 Glary Utilities (Pro) All-in-one system optimization suite. - Registry cleaner - Disk cleaner - Startup manager - Performance boost	24 7-Zip File Manager File manager with built-in compression. 7z, ZIP, RAR, TAR support - High compression ratio - File encryption (256-bit) - Open source

WHAT THEY HELP ACHIEVE

- Stronger Security
- Data Protection & Backup
- System Reliability
- Operational Efficiency
- IT Management & Automation
- Cost Savings
- Compliance & Governance

USED BY ORGANIZATIONS LIKE INVESTEC FOR

- Secure Operations
- Risk Management
- IT Infrastructure Monitoring
- Business Continuity
- Regulatory Compliance
- Productivity & Efficiency

BEST PRACTICES

- Keep software up to date.
- Use strong passwords and encryption.
- Schedule regular backups and test restores.
- Monitor systems and performance regularly.
- Remove unused software and old files.

Utility software — the small tools that keep systems healthy

Cloud computing software (part 1)

- SaaS, PaaS, IaaS — three service models.
- Public, private, hybrid — three deployment models.
- Major clouds: AWS, Azure, Google Cloud.

CLOUD COMPUTING SOFTWARE
Work. Collaborate. Store. Secure. Manage. Anywhere.
Cloud software delivers powerful services that help individuals, teams and organizations be more productive, secure and efficient—anytime, anywhere.

Cloud Platforms (Infrastructure & Services)

- Microsoft Azure:** Build, deploy and manage applications and services through Microsoft's global cloud.
- Amazon Web Services:** Broad and deep cloud platform with the most comprehensive services and global reach.
- Google Cloud Platform:** Secure, data analytics and machine learning solutions that run on the same infrastructure as Google.

Productivity & Collaboration Suites

- Microsoft 365:** Productivity apps and cloud services to help you and your organization achieve more.
- Google Workspace:** Smart, secure and innovative tools to collaborate and get work done.

Cloud Storage & File Sharing

- OneDrive:** Microsoft's cloud storage for files and backups.
- SharePoint:** Team sites, content management and intranet solutions.
- Dropbox:** Secure file storage and sharing made simple.

Management, Identity & Security

- Microsoft Intune:** Cloud based endpoint management to protect and manage devices, apps and data.
- Microsoft Entra ID (Azure AD):** Identity and access management to protect users and resources.

Key Benefits

- Access powerful tools from anywhere, on any device
- Enhance collaboration and communication
- Protect data with enterprise-grade security
- Scale resources up or down in minutes
- Reduce costs and increase operational efficiency
- Stay updated with automatic updates and new features

Why Cloud Software?

- Flexibility:** Adapt quickly to change and scale with ease
- Collaboration:** Work together in real-time from anywhere
- Security:** Protect data with advanced security
- Reliability:** 99.9% uptime and disaster recovery
- Innovation:** Access the latest technology and innovations
- Cost Savings:** Lower IT costs with pay-as-you-go pricing

Cloud software empowers people and organizations to do more—anywhere, securely and efficiently. | Secure. Scalable. Reliable. | The cloud is more than technology—it's a better way to work.

MAJOR CLOUD SERVICE PROVIDERS
Powering the World's Digital Transformation

Provider	Services	Strengths	Best For	Regions
Amazon Web Services	Compute, Storage, Database, Networking, Analytics, AI/ML, IoT, DevOps	Largest global presence, mature services, scalability	Startups to enterprises, highly scalable workloads	34+ regions 100+ Availability Zones
Microsoft Azure	Compute, Storage, Database, AI/ML, Analytics, IoT, DevOps, Hybrid Cloud	Hybrid cloud leader, Microsoft ecosystem, enterprise-ready	Enterprises, .NET apps, Windows workloads	60+ regions 160+ Availability Zones
Google Cloud	Compute, Storage, Database, AI/ML, Data Analytics, Kubernetes, DevOps	Data & AI leader, innovation, open source, enterprise-ready	Data-driven apps, AI/ML, developers, startups	40+ regions 12+ Zones
IBM Cloud	Compute, Storage, Database, AI/ML, Networking, Hybrid Cloud	Hybrid cloud, security, compliance, enterprise solutions	Enterprises, regulated industries	20+ regions 60+ Data Centers
Oracle Cloud	Compute, Storage, Database, Autonomous DB, AI/ML, Networking, Apps	Strong in database, compliance, enterprise apps, performance	Enterprises, Oracle customers	35+ regions 90+ Availability Domains
Alibaba Cloud	Compute, Storage, Database, AI/ML, Big Data, IoT, Networking, Security	Strong in Asia, competitive pricing, e-commerce solutions	Businesses in Asia, e-commerce, startups	25+ regions 86+ Availability Zones
Tencent Cloud	Compute, Storage, Database, AI/ML, Gaming, CDN, Networking, Security	Strong in gaming, social, media, and mobile ecosystem	Gaming, media, internet companies, startups	27+ regions 70+ Availability Zones
DigitalOcean	Compute, Storage, Database, Kubernetes, App Platform, Networking	Simplicity, developer friendly, cost-effective	Developers, startups, SMBs	15+ regions 40+ Data Centers
VMware Cloud	Compute, Storage, Database, Disaster Recovery, Desktop as a Service	Hybrid cloud, VMware ecosystem, migration solutions	VMware customers, hybrid cloud needs	25+ regions Global Data Centers

Global Reach: Hundreds of regions worldwide

High Availability: Built for reliability and uptime

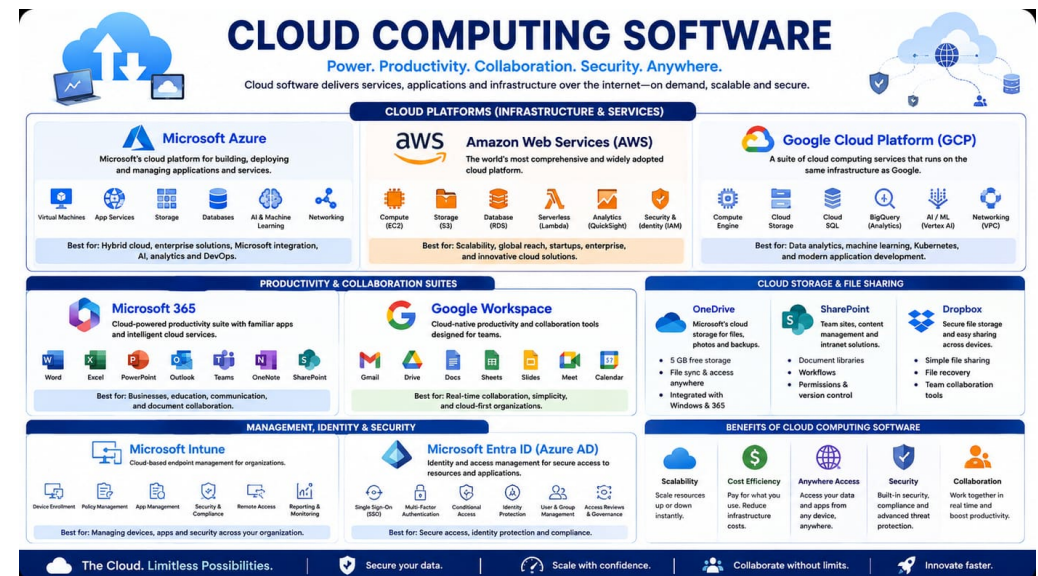
Security First: Advanced security and compliance

Scalable: Scale up or down in minutes

Cost Optimized: Pay for what you use

Cloud computing software (part 2)

- Microsoft 365, Google Workspace: everyday SaaS.
- Cloud storage, cloud backup, cloud identity.
- 'The cloud' = someone else's servers.



Virtualisation and containers

- VMs run whole guest OSes on shared hardware.
- Containers share the host OS, start in seconds.
- Both save cost and enable rapid deployment.

VIRTUALIZATION & CONTAINERS 2026
Powering Modern IT Infrastructure, Development & Cloud

Virtualization and containerization technologies enable you to run more workloads, use resources efficiently, and deliver applications faster and more securely.

WHY IT MATTERS

- Maximize Resource Utilization
- Enable Cloud and Hybrid Environments
- Speed Up Development & Delivery
- Improve Security & Isolation
- Reduce Costs & Complexity

1 Hyper-V
Microsoft's native hypervisor for creating and managing virtual machines on Windows.

Key Features

- Create and manage virtual machines
- Dynamic memory and resource allocation
- Virtual switches and networking
- High availability and failover clustering
- Seamless integration with Windows Server and Azure

2 VMware Workstation
Powerful desktop virtualization software for running multiple operating systems on a single Windows or Linux PC.

Key Features

- Run multiple OS simultaneously
- Snapshots and cloning
- Advanced virtual networking
- 3D graphics acceleration
- Support for a wide range of OS

3 VMware ESXi
Enterprise-grade bare-metal hypervisor for building and managing virtualized data center environments.

Key Features

- High performance and scalability
- Centralized management with vCenter
- Live migration (vMotion)
- High availability and fault tolerance
- Strong security and role-based access

4 Oracle VirtualBox
Open-source desktop virtualization software for running virtual machines across platforms.

Key Features

- Free and open source
- Cross-platform (Windows, Linux, macOS, Solaris)
- Snapshots and cloning
- Shared folders and seamless mode
- Extensive guest OS support

5 Docker
Platform for developing, shipping, and running applications in lightweight containers.

Key Features

- Containerize applications and dependencies
- Consistent environments
- Fast deployment and portability
- Large ecosystem and registry (Docker Hub)
- Easy integration in CI/CD pipelines

6 Kubernetes
Open-source container orchestration platform for automating deployment, scaling, and management of containerized apps.

Key Features

- Automated deployment and scaling
- Self-healing and high availability
- Load balancing and service discovery
- Secret and configuration management
- Extensible with a rich ecosystem

7 Windows Sandbox
Lightweight, isolated desktop environment for safely running applications in a temporary virtual space.

Key Features

- Clean and disposable environment
- Built into Windows 10/11 Pro, Enterprise, Education
- Ideal for testing and running untrusted apps
- No impact on host system
- Quick start and low resource usage

8 Windows Subsystem for Linux (WSL)
Run a Linux environment natively on Windows for development and administration.

Key Features

- Run Linux distributions on Windows
- High performance and system integration
- Access Windows files and Linux tools
- Great for developers and DevOps
- Supports WSL 2 (real Linux kernel)

9 Azure Virtual Machines
Scalable, on-demand virtual machines in the Microsoft Azure cloud platform.

Key Features

- Provision Windows or Linux VMs in minutes
- Global availability and scalability
- Integration with Azure services
- Pay-as-you-go pricing
- Built-in security, monitoring, and backups

VIRTUALIZATION vs CONTAINERS

VIRTUAL MACHINES	CONTAINERS
Full OS virtualization	OS-level virtualization
Heavier and more isolated	Lightweight and portable
Ideal for running different OS	Share host OS kernel
Higher resource usage	Faster startup and efficient
Examples: Hyper-V, VMware, VirtualBox	Examples: Docker, Kubernetes

COMMON USE CASES

- Development & Testing
- Server Consolidation
- Cloud & Hybrid Environments
- CI/CD Pipelines
- Legacy Application Support
- Disaster Recovery & Backup
- Edge & IoT Workloads
- Learning & Training Labs

2026 TRENDS

- Cloud-Native Everything: More workloads moving to containers and serverless platforms.
- Stronger Security & Isolation: Enhanced runtime security, zero trust, and confidential computing.
- AI-Optimized Infrastructure: GPUs and accelerators integrated in VMs and containers.
- Edge & Distributed Computing: Virtualization and containers extending to the edge.

BEST PRACTICES

- Keep your platforms and tools updated.
- Use least privilege access and strong authentication.
- Monitor performance and resource usage continuously.
- Back up your images, VMs, and configurations regularly.
- Automate deployments with infrastructure as code.

☆ Virtualization and containers are the foundation of modern IT. | Run smarter. Scale faster. Secure everything. | 2026 Build the future with virtualized and containerized solutions.

Networking software

- DHCP, DNS, VPN, firewall, load balancer.
- Runs on servers, routers and endpoints.
- Configuration errors here cause most outages.

NETWORKING SOFTWARE 2026
Essential tools to design, configure, manage and troubleshoot modern networks.
From network simulation and device configuration to secure remote access and file transfer, these tools help IT professionals build, manage and secure connectivity.

NETWORKING SOFTWARE HELPS YOU

- Design & Simulate
- Configure & Manage
- Monitor & Analyze
- Secure & Protect
- Connect & Access
- Transfer & Share

Tool	Description	Key Features
1 Windows Network Manager	Built-in Windows tool for viewing and managing network connections, status, and settings.	- View network status and properties - Configure IP, DNS, and adapter settings - Manage Wi-Fi and Ethernet connections - Troubleshoot connectivity issues - Monitor network performance
2 pfSense	Open-source firewall and router platform based on FreeBSD, ideal for network security and management.	- Stateful firewall and NAT - VPN (Pfsense, OpenVPN, WireGuard) - Intrusion detection & prevention (IDS) - Traffic shaping and bandwidth control - High availability and failover
3 Cisco Packet Tracer	Network simulation and visualization tool for learning, teaching, and prototyping network designs.	- Simulate complex network topologies - Configure routers, switches, PCs, etc. - Supports real Cisco CLI commands - Visualize protocols and traffic flow - Ideal for training and certification
4 Cisco IOS	Cisco Internetwork Operating System that powers Cisco routers and switches.	- Reliable and secure network OS - Wide range of routing protocols - Scalability and high performance - Advanced security features - Extensive CLI configuration
5 WinBox (MikroTik)	Configuration tool for managing MikroTik RouterOS devices graphically.	- Full access to RouterOS settings - User-friendly GUI - Manage users, firewall, routes, wireless and more - Supports remote management
6 PuTTY	Lightweight SSH and telnet client for secure remote command-line access.	- SSH, Telnet, rlogin, raw socket - Save and manage session profiles - Lightweight and fast - Secure remote access - Cross-platform (Windows)
7 OpenVPN	Open-source VPN solution for creating secure point-to-point or site-to-site connections.	- SSL/TLS encryption - Remote access and site-to-site VPN - Cross-platform support - Flexible authentication options - High security and performance
8 WireGuard	Modern, fast, and secure VPN software with a lightweight codebase.	- High performance and low latency - Strong encryption (ChaCha20, Curve25519) - Small and easy to audit - Simple configuration - Cross-platform support
9 FileZilla	Popular FTP, FTPS and SFTP client for transferring files between local and remote hosts.	- Supports FTP, FTPS, SFTP - Drag-and-drop interface - Site Manager for easy access - Transfer queue and resume support - Cross-platform (Windows, macOS, Linux)
10 Remote Desktop	Microsoft's remote access tool for connecting to and controlling remote computers.	- Remote access to Windows PCs - File transfer and clipboard sharing - Multi-monitor support - Secure with RLA and encryption - Easy to use and built-in (Windows)

COMMON NETWORKING USE CASES

- Network Design & Simulation:** Plan and simulate networks before deployment.
- Network Security & VPN:** Secure networks and enable safe remote connections.
- Device Configuration & Management:** Configure routers, switches and firewalls efficiently.
- Monitoring & Troubleshooting:** Monitor performance and resolve network issues.
- File Transfer & Sharing:** Transfer files securely across the network.
- Remote Access & Administration:** Access and manage systems from anywhere securely.

DID YOU KNOW?
Modern networks rely on a combination of tools for design, security, management, and access. Using the right software improves efficiency, security, and uptime.

NETWORKING BEST PRACTICES

- Keep software updated for security and features
- Use strong authentication and encryption
- Regularly back up configurations
- Monitor network traffic and logs
- Document network designs and changes

2026 NETWORKING TRENDS

- Cloud Networking:** More tools integrate with cloud platforms and hybrid networks.
- Zero Trust Networking:** Secure access is the new standard.
- AI-Powered Monitoring:** AI helps detect issues faster and predict problems.
- Software-Defined Networking (SDN):** More automation and centralized network control.

TIPS FOR GETTING STARTED

- Start with one tool and master the basics
- Practice in a virtual lab environment
- Use realistic scenarios and document your work
- Join online communities and forums
- Keep learning and stay updated

Stay Connected. Stay Secure. Stay Productive. | Build. Manage. Secure. Optimize. | The right tools. The right connections. The right future. **2026**

Cybersecurity software

- Antivirus, EDR, SIEM, MFA, encryption.
- Defence in depth — no single product is enough.
- Users are still the biggest attack surface.

CYBERSECURITY SOFTWARE
The 2026 Cybersecurity Toolkit
Powerful tools that protect, detect, respond, and recover in an ever-evolving digital world.

2026 EDITION

Protect Prevent threats | Detect Find threats early | Respond Act fast & contain | Recover Restore & strengthen | Comply Meet standards & regulations

ENDPOINT PROTECTION	NETWORK & INFRASTRUCTURE SECURITY	CLOUD SECURITY	APPLICATION & WEB SECURITY	SECURITY OPERATIONS & MONITORING	IDENTITY & ACCESS MANAGEMENT
Microsoft Defender for Endpoint Advanced threat protection for Windows, macOS, Linux, iOS & Android. CrowdStrike Falcon AI-powered endpoint protection & EDR. Bitdefender Next-gen antivirus with ransomware protection. ESET Lightweight, fast, effective endpoint security. Kaspersky Real-time protection against cyber threats.	Cisco Secure Comprehensive network & cloud security solutions. Fortinet FortiGate Firewall, VPN, & threat protection for networks. Palo Alto Networks Cortex Advanced network security & threat intelligence. Check Point Infinity Prevent threats across networks, cloud & users. Sophos XGS Firewall Next-gen firewall with AI-driven protection.	Microsoft Entra ID Protection Secure identities, access & cloud resources. Google Cloud Security Command Center Threat detection & risk management in GCP. AWS AWS GuardDuty Intelligent threat detection for AWS workloads. Cloudflare Protect websites, APIs & networks (DDoS, WAF, CDN). SailPoint Identity security & access governance.	OWASP ZAP Web app vulnerability scanner (open source). Burp Suite Professional Web security testing & vulnerability scanner. Veracode Secure code analysis & application security. Akamai App & API Protector Protect APIs & web apps from attacks. Sucuri SiteCheck Website malware scanning & firewall.	Splunk Enterprise Security Security monitoring, analytics & SIEM. Microsoft Sentinel Cloud-native SIEM & SOAR solution. IBM QRadar Threat detection, investigation & response. Elastic Security Search, analyze & respond to threats. AlienVault OSSIM Open source SIEM & threat detection.	Okta Identity management & single sign-on (SSO). Ping Identity Secure digital identity solutions. Duo Security Multi-factor authentication & zero-trust access. CyberArk Privileged access management. LastPass Secure password management.
Threat Intelligence & Analysis Recorded Future Real-time threat intelligence. ThreatConnect Intelligence-driven threat response. VirusTotal Analyze suspicious files, URLs & IPs. MISP Threat information sharing platform.	Proofpoint Advanced email threat protection. Barracuda Email Security Gateway Stop phishing, spam & malware. Abnormal Security AI-powered email threat detection. Mimecast Cloud email security & continuity.	Data Protection & Encryption Acronis Cyber Protect Backup, ransomware protection & recovery. Boxcryptor Easy encryption for cloud storage. SQL Server TDE (Transparent Data Encryption) Encrypt sensitive data at database level.	Mobile Security Lookout Mobile threat protection for iOS & Android. Avast Mobile Security Protects phones from malware & phishing. Norton Mobile Security Advanced mobile threat detection. Zimperium Mobile device & app security.	Offensive Security & Testing Kali Linux Penetration testing & ethical hacking. Metasploit Framework Exploit development & penetration testing. Nmap Network discovery & security auditing. Aircrack-ng Wireless network security auditing.	Backup Recovery & Business Continuity Veeam Backup & Replication Reliable backups & quick recovery. Acronis Cyber Protect All-in-one protection (backup + security). Carbonite Cloud backup for individuals & businesses. Dell PowerProtect Data Manager Data protection, backup & recovery.
COMPLIANCE & GOVERNANCE Tenable.io Vulnerability management & compliance. Qualys Security, compliance & asset management. OneTrust Data privacy, consent & compliance platform.					

DID YOU KNOW?
Cyber threats are evolving every day. These tools help you stay one step ahead, protect vital resources, and build a more secure future.

CYBERSECURITY BEST PRACTICES
 Stay informed. Stay Protected. Stay Secure. | Cybersecurity is everyone's responsibility. | Protect your data. Protect your future.

EMERGING TRENDS IN 2026
 AI-powered threat detection
 Zero Trust architecture
 Ransomware-as-a-Service (RaaS) countermeasures
 Cloud-native security solutions
 Automation & SOAR platforms

- ## AI software

